Solutions to Axler's Linear Algebra

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Contents

1 Vector Spaces

1A $\mathbb{R}^n$ and $\mathbb{C}^n$

Exercise 1A1

Show that $\alpha + \beta = \beta + \alpha$ for all $\alpha, \beta \in \mathbb{C}$.

Solution. Let $\alpha = a + bi$ and $\beta = c + di$ for some $a, b, c, d \in \mathbb{R}$. Then

$$\begin{aligned} \alpha + \beta &= (a + bi) + (c + di) \\ &= (a + c) + (b + d)i \\ &= (c + a) + (d + b)i \\ &= (c + di) + (a + bi) = \beta + \alpha. \end{aligned}$$

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Exercise 1A2

Show that $(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma)$ for all $\alpha, \beta, \gamma \in \mathbb{C}$.

Solution. Let $\alpha = a + bi$, $\beta = c + di$, and $\gamma = e + fi$ for some $a, b, c, d, e, f \in \mathbb{R}$. Then

$$\begin{aligned} (\alpha + \beta) + \gamma &= ((a + bi) + (c + di)) + (e + fi) \\ &= ((a + c) + (b + d)i) + (e + fi) \\ &= (a + c + e) + (b + d + f)i \\ &= (a + (c + e)) + (b + (d + f))i \\ &= (a + bi) + ((c + di) + (e + fi)) = \alpha + (\beta + \gamma). \end{aligned}$$

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Exercise 1A3

Show that $(\alpha\beta)\gamma = \alpha(\beta\gamma)$ for all $\alpha, \beta, \gamma \in \mathbb{C}$.

Solution. Let $\alpha = a + bi$, $\beta = c + di$, and $\gamma = e + fi$ for some $a, b, c, d, e, f \in \mathbb{R}$. Then

$$\begin{aligned} (\alpha\beta)\gamma &= ((a + bi)(c + di))(e + fi) \\ &= ((ac - bd) + (ad + bc)i)(e + fi) \\ &= ((ac - bd)e - (ad + bc)f) + ((ac - bd)f + (ad + bc)e)i \\ &= (a(ce) - b(de) - a(df) - b(cf)) + (a(cf) - b(df) + a(de) + b(ce))i \\ &= \alpha(\beta\gamma). \end{aligned}$$

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Exercise 1A4

Show that $\gamma(\alpha + \beta) = \gamma\alpha + \gamma\beta$ for all $\alpha, \beta, \gamma \in \mathbb{C}$.

Solution. Let $\alpha = a + bi$, $\beta = c + di$, and $\gamma = e + fi$ for some $a, b, c, d, e, f \in \mathbb{R}$. Then

$$\begin{aligned}
\gamma(\alpha+\beta) &= (e+fi)((a+bi)+(c+di)) \\
&= (e+fi)((a+c)+(b+d)i) \\
&= (e(a+c)-f(b+d)) + (e(b+d)+f(a+c))i \\
&= (ea - fb + ec - fd) + (eb + fa + ed + fc)i \\
&= (ea - fb) + (eb + fa)i + (ec - fd) + (ed + fc)i \\
&= (e+fi)(a+bi) + (e+fi)(c+di) = \gamma\alpha + \gamma\beta.
\end{aligned}$$

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Exercise 1A5

Show that for every $\alpha \in \mathbb{C}$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha + \beta = 0$.

Solution. Let $\alpha = a + bi$ for some $a, b \in \mathbb{R}$. We want to find $\beta = c + di$ such that $\alpha + \beta = 0$. This gives

$$(a+bi) + (c+di) = (a+c) + (b+d)i = 0 + 0i.$$

Solving for c and d, we get $c = -a$ and $d = -b$. Hence, $\beta = c + di = -a - bi = -\alpha$.

To see that β is unique, suppose there is another $\gamma \in \mathbb{C}$ such that $\alpha + \gamma = 0$. Then

$$\beta = \beta + 0 = \beta + (\alpha + \gamma) = (\beta + \alpha) + \gamma = 0 + \gamma = \gamma.$$

Hence, β is unique. ■

Exercise 1A6

Show that for every $\alpha \in \mathbb{C}$ with $\alpha \neq 0$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha\beta = 1$.

Solution. Let $\alpha = a + bi$ for some $a, b \in \mathbb{R}$ with $\alpha \neq 0$. We want to find $\beta = c + di$ such that $\alpha\beta = 1$. This gives

$$(a+bi)(c+di) = (ac - bd) + (ad + bc)i = 1 + 0i.$$

Solving for c and d, we get the system of equations

$$\begin{cases} ac - bd = 1 \\ ad + bc = 0 \end{cases} \implies \begin{cases} a^2c - abd = a \\ abd + b^2c = 0 \end{cases}$$

Adding the two equations, we get

$$(a^2 + b^2)c = a \implies c = \frac{a}{a^2+b^2}.$$

Substituting this into the second equation, we get

$$abd + b^2\frac{a}{a^2+b^2} = 0 \implies d = \frac{-b}{a^2+b^2}.$$

Hence,

$$\beta = c + di = \frac{a}{a^2+b^2} + \frac{-b}{a^2+b^2}i.$$

If $\gamma \in \mathbb{C}$ is another complex number such that $\alpha\gamma = 1$, then

$$\gamma = \gamma 1 = \gamma(\alpha\beta) = (\gamma\alpha)\beta = 1\beta = \beta.$$

Hence, β is unique. ■

Exercise 1A7

Show that
$$\frac{-1+\sqrt{3}i}{2}$$
is a cube root of 1.

Solution. Note that the complex number given can be written as
$$\frac{1}{2}(-1+\sqrt{3}i).$$
Cubing $-1+\sqrt{3}i$, we get
$$\begin{aligned}(-1+\sqrt{3}i)^3 &= (-1)^3 + 3(-1)^2(\sqrt{3}i) + 3(-1)(\sqrt{3}i)^2 + (\sqrt{3}i)^3\\ &= -1 + 3\sqrt{3}i + 3(-1)(-3) + (\sqrt{3}i)^3\\ &= -1 + 3\sqrt{3}i + 9 + 3\sqrt{3}i(-i)\\ &= -1 + 3\sqrt{3}i + 9 - 3\sqrt{3}i\\ &= 8.\end{aligned}$$
Therefore,
$$\left(\frac{-1+\sqrt{3}i}{2}\right)^3 = \frac{(-1+\sqrt{3}i)^3}{8} = \frac{8}{8} = 1.$$
■

Exercise 1A8

Find two distinct square roots of i.

Solution. Let $z = a + bi$ for some $a, b \in \mathbb{R}$. We want to find z such that $z^2 = i$. This gives
$$(a+bi)^2 = a^2 - b^2 + 2abi = 0 + 1i.$$
Solving for a and b, we get the system of equations
$$\begin{cases} a^2 - b^2 = 0\\ 2ab = 1\end{cases}$$
The first equation yields $a = b$ or $a = -b$. If $a = b$, then the second equation gives
$$2b^2 = 1 \implies b = \pm\frac{1}{\sqrt{2}} \implies a = \pm\frac{1}{\sqrt{2}}.$$
If $a = -b$, this results in $-2b^2 = 1$ which has no solution when $b \in \mathbb{R}$. Hence, the two distinct square roots of i are
$$\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \quad \text{and} \quad -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i.$$
■

Exercise 1A9

Find $x \in \mathbb{R}^4$ such that
$$(4,-3,1,7) + 2x = (5,9,-6,8).$$

Solution. Let $x = (x_1, x_2, x_3, x_4) \in \mathbb{R}^4$. The equation becomes

$$(4, -3, 1, 7) + 2(x_1, x_2, x_3, x_4) = (5, 9, -6, 8),$$

which simplifies to the system of equations

$$\begin{cases} 5 = 4 + 2x_1 \\ 9 = -3 + 2x_2 \\ -6 = 1 + 2x_3 \\ 8 = 7 + 2x_4 \end{cases} \implies \begin{cases} 2x_1 = 1 \\ 2x_2 = 12 \\ 2x_3 = -7 \\ 2x_4 = 1 \end{cases}$$

Therefore,

$$x = \left(\frac{1}{2}, 6, -\frac{7}{2}, \frac{1}{2}\right).$$

■

Exercise 1A10

Explain why there does not exist $\lambda \in \mathbb{C}$ such that

$$\lambda(2 - 3i, 5 + 4i, -6 + 7i) = (12 - 5i, 7 + 22i, -32 - 9i).$$

Solution. Suppose there exists $\lambda \in \mathbb{C}$ such that

$$\lambda(2 - 3i, 5 + 4i, -6 + 7i) = (12 - 5i, 7 + 22i, -32 - 9i).$$

This gives the system of equations

$$\begin{cases} \lambda(2 - 3i) = 12 - 5i \\ \lambda(5 + 4i) = 7 + 22i \\ \lambda(-6 + 7i) = -32 - 9i \end{cases}$$

Solving the first equation for λ, we get

$$\lambda = \frac{12 - 5i}{2 - 3i} = \frac{(12 - 5i)(2 + 3i)}{(2 - 3i)(2 + 3i)} = \frac{24 + 36i - 10i - 15i^2}{4 + 9} = \frac{24 + 26i + 15}{13} = 3 + 2i.$$

Solving the second equation for λ, we get

$$\lambda = \frac{7 + 22i}{5 + 4i} = \frac{(7 + 22i)(5 - 4i)}{(5 + 4i)(5 - 4i)} = \frac{35 - 28i + 110i - 88i^2}{25 + 16} = \frac{35 + 82i + 88}{41} = 3 + 2i.$$

Solving the third equation for λ, we get

$$\lambda = \frac{-32 - 9i}{-6 + 7i} = \frac{(-32 - 9i)(-6 - 7i)}{(-6 + 7i)(-6 - 7i)} = \frac{192 + 224i + 54i + 63i^2}{36 + 49} = \frac{192 + 278i - 63}{85} = \frac{129 + 278i}{85}.$$

Since the value of λ obtained from the third equation is different from the value obtained from the first two equations, no such $\lambda \in \mathbb{C}$ exists. ■

Exercise 1A11

Show that $(x + y) + z = x + (y + z)$ for all $x, y, z \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \dots, x_n)$, $y = (y_1, y_2, \dots, y_n)$, and $z = (z_1, z_2, \dots, z_n)$. Then

$$\begin{aligned}(x+y)+z &= ((x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n)) + (z_1, z_2, \dots, z_n) \\ &= ((x_1+y_1), (x_2+y_2), \dots, (x_n+y_n)) + (z_1, z_2, \dots, z_n) \\ &= (x_1+y_1+z_1, x_2+y_2+z_2, \dots, x_n+y_n+z_n) \\ &= (x_1, x_2, \dots, x_n) + ((y_1+z_1), (y_2+z_2), \dots, (y_n+z_n)) \\ &= x + (y+z).\end{aligned}$$

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Exercise 1A12

Show that $(ab)x = a(bx)$ for all $a, b \in \mathbb{F}$ and $x \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \dots, x_n) \in \mathbb{F}^n$ and $a, b \in \mathbb{F}$. Then

$$\begin{aligned}(ab)x &= (ab)(x_1, x_2, \dots, x_n) \\ &= ((ab)x_1, (ab)x_2, \dots, (ab)x_n) \\ &= (a(bx_1), a(bx_2), \dots, a(bx_n)) \\ &= a((bx_1, bx_2, \dots, bx_n)) \\ &= a(bx).\end{aligned}$$

■

Exercise 1A13

Show that $1x = x$ for all $x \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \dots, x_n) \in \mathbb{F}^n$. Then

$$\begin{aligned}1x &= 1(x_1, x_2, \dots, x_n) \\ &= (1x_1, 1x_2, \dots, 1x_n) \\ &= (x_1, x_2, \dots, x_n) \\ &= x.\end{aligned}$$

■

Exercise 1A14

Show that $\lambda(x+y) = \lambda x + \lambda y$ for all $\lambda \in \mathbb{F}$ and $x, y \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \dots, x_n)$ and $y = (y_1, y_2, \dots, y_n)$ be elements of $\mathbb{F}^n$ and $\lambda \in \mathbb{F}$. Then

$$\begin{aligned}\lambda(x+y) &= \lambda((x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n)) \\ &= \lambda((x_1+y_1), (x_2+y_2), \dots, (x_n+y_n)) \\ &= (\lambda(x_1+y_1), \lambda(x_2+y_2), \dots, \lambda(x_n+y_n)) \\ &= (\lambda x_1 + \lambda y_1, \lambda x_2 + \lambda y_2, \dots, \lambda x_n + \lambda y_n) \\ &= (\lambda x_1, \lambda x_2, \dots, \lambda x_n) + (\lambda y_1, \lambda y_2, \dots, \lambda y_n) \\ &= \lambda(x_1, x_2, \dots, x_n) + \lambda(y_1, y_2, \dots, y_n) \\ &= \lambda x + \lambda y.\end{aligned}$$

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Exercise 1A15

Show that $(a+b)x = ax + bx$ for all $a, b \in \mathbb{F}$ and $x \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \dots, x_n) \in \mathbb{F}^n$ and $a, b \in \mathbb{F}$. Then

$$\begin{aligned}(a+b)x &= (a+b)(x_1, x_2, \dots, x_n)\\ &= ((a+b)x_1, (a+b)x_2, \dots, (a+b)x_n)\\ &= (ax_1 + bx_1, ax_2 + bx_2, \dots, ax_n + bx_n)\\ &= (ax_1, ax_2, \dots, ax_n) + (bx_1, bx_2, \dots, bx_n)\\ &= ax + bx.\end{aligned}$$

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1B Definition of Vector Space

Exercise 1B1

Prove that $-(-v) = v$ for every $v \in V$.

Solution. Let $v \in V$. By the definition of additive inverse, we have

$$v + (-v) = 0 = -(-v) + (-v).$$

Adding $-(-v)$ to both sides, we get

$$v + (-v) + (-(-v)) = -(-v) + (-v) + (-(-v)).$$

Simplifying both sides, we obtain

$$v = -(-v).$$

■

Exercise 1B2

Suppose $a \in \mathbb{F}$, $v \in V$, and $av = 0$. Prove that $a = 0$ or $v = 0$.

Solution. If $a = 0$, then we are done. Suppose $a \neq 0$. Then

$$v = 1v = \frac{1}{a}(av) = \frac{1}{a} \cdot 0 = 0.$$

■

Exercise 1B3

Suppose $v, w \in V$. Explain why there exists a unique $x \in V$ such that $v + 3x = w$.

Solution. Solving for x explicitly, we obtain

$$3x = w - v \implies x = \frac{1}{3}(w - v).$$

This shows that there exists $x \in V$ such that $v + 3x = w$. To see that x is unique, suppose there exists another $y \in V$ such that $v + 3y = w$. Then dividing by 3, we get

$$3x = w - v = 3y \implies x = y.$$

■

Exercise 1B4

The empty set is not a vector space. The empty set fails to satisfy only one of the requirements listed in the definition of a vector space. Which one?

Solution. The empty set fails to satisfy the requirement that there exists a zero vector $0 \in V$ such that $v + 0 = v$ for all $v \in V$ since it contains no elements. ■

Exercise 1B5

Show that in the definition of a vector space, the additive inverse condition can be replaced with the condition that

$$0v = 0 \text{ for all } v \in V.$$

Here, the 0 on the left side is the number 0, and the 0 on the right side is the additive identity in V.

Solution. Suppose that $0v = 0$ for all $v \in V$. Let $v \in V$. Then

$$v + (-1)v = 1v + (-1)v = (1 + (-1))v = 0v = 0.$$

Hence, $(-1)v$ is the additive inverse of v.

Conversely, suppose that for every $v \in V$, there exists an additive inverse $-v \in V$ such that $v + (-v) = 0$. Then $0v = (0 + 0)v = 0v + 0v$. Adding $-0v$ to both sides, we get $0 = 0v$. Hence, $0v = 0$ for all $v \in V$. ■

Exercise 1B6

Let ∞ and $-\infty$ denote two distinct objects, neither of which is in $\mathbb{R}$. Define an addition and scalar multiplication on $\mathbb{R} \cup \{\infty, -\infty\}$ as you could guess from the notation. Specifically, the sum and product of two real numbers is as usual, and for $t \in \mathbb{R}$, define

$$t\infty = \begin{cases} -\infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ \infty & \text{if } t > 0, \end{cases} \quad t(-\infty) = \begin{cases} \infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ -\infty & \text{if } t > 0, \end{cases}$$

and

$$\begin{aligned} t + \infty &= \infty + t = \infty + \infty = \infty, \\ t + (-\infty) &= (-\infty) + t = (-\infty) + (-\infty) = -\infty, \\ \infty + (-\infty) &= (-\infty) + \infty = 0. \end{aligned}$$

With these operations of addition and scalar multiplication, is $\mathbb{R} \cup \{\infty, -\infty\}$ a vector space over $\mathbb{R}$? Explain.

Solution. No, $\mathbb{R} \cup \{\infty, -\infty\}$ is not a vector space over $\mathbb{R}$ as it does not satisfy the associative property of addition. Let $t \in \mathbb{R}$. Then

$$(t + \infty) + (-\infty) = \infty + (-\infty) = 0 \neq t = t + 0 = t + (\infty + (-\infty)).$$ ■

Exercise 1B7

Suppose S is a nonempty set. Let V^S denote the set of functions from S to V. Define a natural addition and scalar multiplication on V^S, and show that V^S is a vector space with these definitions.

Solution. Let $f, g \in V^S$ and $t \in \mathbb{F}$. Define addition and scalar multiplication as follows:

$$(f + g)(s) = f(s) + g(s), \quad (tf)(s) = t(f(s))$$

for all $s \in S$. We will show that V^S is a vector space with these operations.

1. **Commutativity**: Let $f, g \in V^S$. Then for all $s \in S$,

$$(f + g)(s) = f(s) + g(s) = g(s) + f(s) = (g + f)(s).$$

2. **Associativity**: Let $f, g, h \in V^S$ and $t, u \in \mathbb{F}$. Then for all $s \in S$,

$$\begin{aligned} ((f + g) + h)(s) &= (f + g)(s) + h(s) \\ &= (f(s) + g(s)) + h(s) \\ &= f(s) + (g(s) + h(s)) \\ &= f(s) + (g + h)(s) \\ &= (f + (g + h))(s), \end{aligned}$$

and

$$
\begin{aligned}
((tu)f)(s) &= (tu)(f(s)) \\
&= t(u(f(s))) \\
&= t(uf)(s) \\
&= (t(uf))(s).
\end{aligned}
$$

3. **Additive identity**: Let $0 \in V$ be the additive identity in V. Define the zero function $0_S \in V^S$ by $0_S(s) = 0$ for all $s \in S$. Then for all $f \in V^S$ and $s \in S$,

$$(f + 0_S)(s) = f(s) + 0_S(s) = f(s) + 0 = f(s) = 0 + f(s) = (0_S + f)(s).$$

Hence, 0_S is the additive identity in V^S.

4. **Additive inverse**: Let $f \in V^S$. Define the function $-f \in V^S$ by $(-f)(s) = -f(s)$ for all $s \in S$. Then for all $s \in S$,

$$(f + (-f))(s) = f(s) + (-f)(s) = f(s) + (-f(s)) = 0 = 0_S(s).$$

Hence, $-f$ is the additive inverse of f in V^S.

5. **Multiplicative identity**: For all $f \in V^S$ and $s \in S$,

$$(1f)(s) = 1(f(s)) = f(s).$$

6. **Distributive properties**: Let $f, g \in V^S$ and $t, u \in \mathbb{F}$. Then for all $s \in S$,

$$
\begin{aligned}
((t+u)f)(s) &= (t+u)(f(s)) \\
&= t(f(s)) + u(f(s)) \\
&= (tf)(s) + (uf)(s) \\
&= (tf + uf)(s),
\end{aligned}
$$

and

$$
\begin{aligned}
(t(f+g))(s) &= t((f+g)(s)) \\
&= t(f(s) + g(s)) \\
&= t(f(s)) + t(g(s)) \\
&= (tf)(s) + (tg)(s) \\
&= (tf + tg)(s).
\end{aligned}
$$

■

Exercise 1B8

Suppose V is a real vector space.

- The **complexification** of V, denoted by $V_{\mathbb{C}}$, equals $V \times V$. An element of $V_{\mathbb{C}}$ is an ordered pair (u, v), where $u, v \in V$, but we write this as $u + iv$.
- Addition on $V_{\mathbb{C}}$ is defined by
$$(u_1 + iv_1) + (u_2 + iv_2) = (u_1 + u_2) + i(v_1 + v_2)$$
for all $u_1, u_2, v_1, v_2 \in V$.
- Complex scalar multiplication on $V_{\mathbb{C}}$ is defined by
$$(a + ib)(u + iv) = (au - bv) + i(av + bu)$$
for all $a, b \in \mathbb{R}$ and $u, v \in V$.

Prove that with the definitions of addition and scalar multiplication as above, $V_{\mathbb{C}}$ is a complex vector space.

Solution. We prove that $V_{\mathbb{C}}$ is a complex vector space by verifying the vector space axioms.

1. **Commutativity**: Let $u_1 + v_1 i, u_2 + v_2 i \in V_{\mathbb{C}}$. Then
$$(u_1 + v_1 i) + (u_2 + v_2 i) = (u_1 + u_2) + i(v_1 + v_2) = (u_2 + u_1) + i(v_2 + v_1) = (u_2 + v_2 i) + (u_1 + v_1 i).$$

2. **Associativity**: Let $u_1 + v_1 i, u_2 + v_2 i, u_3 + v_3 i \in V_{\mathbb{C}}$. Then
$$\begin{aligned}((u_1 + v_1 i) + (u_2 + v_2 i)) + (u_3 + v_3 i) &= ((u_1 + u_2) + i(v_1 + v_2)) + (u_3 + v_3 i) \\ &= ((u_1 + u_2) + u_3) + i((v_1 + v_2) + v_3) \\ &= (u_1 + (u_2 + u_3)) + i(v_1 + (v_2 + v_3)) \\ &= (u_1 + v_1 i) + ((u_2 + v_2 i) + (u_3 + v_3 i)).\end{aligned}$$
Similarly, let $a + bi, c + di \in \mathbb{C}$ and $u + vi \in V_{\mathbb{C}}$. Then
$$\begin{aligned}((a + bi)(c + di))(u + vi) &= ((ac - bd) + i(ad + bc))(u + vi) \\ &= ((ac - bd)u - (ad + bc)v) + i((ac - bd)v + (ad + bc)u) \\ &= (a(cu - dv) - b(du + cv)) + i(a(du + cv) + b(cu - dv)) \\ &= (a + bi)((cu - dv) + i(du + cv)) \\ &= (a + bi)((c + di)(u + vi)).\end{aligned}$$

3. **Additive identity**: Let $0 \in V$ be the additive identity in V. Then for all $u + vi \in V_{\mathbb{C}}$,
$$(u + vi) + (0 + 0i) = (u + 0) + i(v + 0) = u + vi = (0 + 0i) + (u + vi).$$
Hence, $0 + 0i$ is the additive identity in $V_{\mathbb{C}}$.

4. **Additive inverse**: Let $u + vi \in V_{\mathbb{C}}$. Then
$$(u + vi) + (-u + (-v)i) = (u - u) + i(v - v) = 0 + 0i.$$
Hence, $-u + (-v)i$ is the additive inverse of $u + vi$ in $V_{\mathbb{C}}$.

5. **Multiplicative identity**: For all $u + vi \in V_{\mathbb{C}}$,

$$(1 + 0i)(u + vi) = (1u - 0v) + i(1v + 0u) = u + vi.$$

Hence, $1 + 0i$ is the multiplicative identity in $V_{\mathbb{C}}$.

6. **Distributive properties**: Let $a + bi, c + di \in \mathbb{C}$ and $u_1 + v_1 i, u_2 + v_2 i \in V_{\mathbb{C}}$. Then

$$\begin{aligned}
(a + bi)((u_1 + v_1 i) + (u_2 + v_2 i)) &= (a + bi)((u_1 + u_2) + i(v_1 + v_2)) \\
&= (a(u_1 + u_2) - b(v_1 + v_2)) + i(a(v_1 + v_2) + b(u_1 + u_2)) \\
&= (au_1 - bv_1) + i(av_1 + bu_1) + (au_2 - bv_2) + i(av_2 + bu_2) \\
&= (a + bi)(u_1 + v_1 i) + (a + bi)(u_2 + v_2 i),
\end{aligned}$$

and

$$\begin{aligned}
((a + bi) + (c + di))(u_1 + v_1 i) &= ((a + c) + (b + d)i)(u_1 + v_1 i) \\
&= ((a + c)u_1 - (b + d)v_1) + i((a + c)v_1 + (b + d)u_1) \\
&= (au_1 - bv_1) + i(av_1 + bu_1) + (cu_1 - dv_1) + i(cv_1 + du_1) \\
&= (a + bi)(u_1 + v_1 i) + (c + di)(u_1 + v_1 i).
\end{aligned}$$

■

1C Subspaces

Exercise 1C1

For each of the following subsets of $\mathbb{F}^3$, determine whether it is a subspace of $\mathbb{F}^3$.

(a) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 \mid x_1 + 2x_2 + 3x_3 = 0\}$

(b) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 \mid x_1 + 2x_2 + 3x_3 = 4\}$

(c) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 \mid x_1 x_2 x_3 = 0\}$

(d) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 \mid x_1 = 5x_3\}$

Solution. Let U be the subset of $\mathbb{F}^3$ in question.

(a) Since the zero vector $(0, 0, 0)$ satisfies $0 + 2(0) + 3(0) = 0$, U is nonempty. Let $u = (u_1, u_2, u_3)$ and $v = (v_1, v_2, v_3)$ be elements of U. Then

$$(u_1 + v_1) + 2(u_2 + v_2) + 3(u_3 + v_3) = (u_1 + 2u_2 + 3u_3) + (v_1 + 2v_2 + 3v_3) = 0 + 0 = 0.$$

Hence, $u + v$ is in U. Let $a \in \mathbb{F}$. Then

$$au_1 + 2(au_2) + 3(au_3) = a(u_1 + 2u_2 + 3u_3) = a \cdot 0 = 0.$$

Hence, au is in U. Therefore, U is a subspace of $\mathbb{F}^3$.

(b) The zero vector $(0, 0, 0)$ does not satisfy $0 + 2(0) + 3(0) = 4$, so U does not contain the zero vector. Hence, it is not a subspace of $\mathbb{F}^3$.

(c) Suppose $u = (1, 0, 0)$ and $v = (0, 1, 1)$. Then u and v are in U since $1 \cdot 0 \cdot 0 = 0$ and $0 \cdot 1 \cdot 1 = 0$. However, $u + v = (1, 1, 1)$ is not in U since $1 \cdot 1 \cdot 1 = 1 \neq 0$. Hence, U is not closed under addition and is not a subspace of $\mathbb{F}^3$.

(d) Since the zero vector $(0, 0, 0)$ satisfies $0 = 5(0)$, U is nonempty. Let $u = (u_1, u_2, u_3)$ and $v = (v_1, v_2, v_3)$ be elements of U. Then

$$u_1 + v_1 = 5u_3 + 5v_3 = 5(u_3 + v_3).$$

Hence, $u + v$ is in U. Let $a \in \mathbb{F}$. Then

$$au_1 = a(5u_3) = 5(au_3).$$

Hence, au is in U. Therefore, U is a subspace of $\mathbb{F}^3$. ■

Exercise 1C2

Verify all assertions about subspaces in Example 1.35.

(a) If $b \in \mathbb{F}$, then

$$\{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 \mid x_3 = 5x_4 + b\}$$

is a subspace of $\mathbb{F}^4$ if and only if $b = 0$.

(b) The set of continuous, real-valued functions on the interval $[0, 1]$ is a subspace of $\mathbb{R}^{[0,1]}$.

(c) The set of differentiable real-valued functions on $\mathbb{R}$ is a subspace of $\mathbb{R}^{\mathbb{R}}$.

(d) The set of differentiable real-valued functions f on the interval $(0, 3)$ such that $f'(2) = b$ is a subspace of $\mathbb{R}^{(0,3)}$ if and only if $b = 0$.

(e) The set of all sequences of complex numbers with limit 0 is a subspace of $\mathbb{C}^\infty$.

Solution. Let U be the subset of $\mathbb{F}^4$ in question.

(a) Suppose U is a subspace of $\mathbb{F}^4$. Then it must contain the zero vector $(0, 0, 0, 0)$. Then $0 = 5(0) + b$, which implies $b = 0$.

Conversely, suppose $b = 0$. Then $U = \{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 \mid x_3 = 5x_4\}$. The proof is similar to that of Exercise 1C1 (d), hence U is a subspace of $\mathbb{F}^4$.

(b) Note that the 0 function on $[0, 1]$ is continuous, so $0 \in U$. Let $f, g \in U$ be continuous functions. Then $f + g$ is continuous, so $f + g \in U$. Let $a \in \mathbb{R}$. Then af is continuous, so $af \in U$. Therefore, U is a subspace of $\mathbb{R}^{[0,1]}$.

(c) Since the derivative of the 0 function is the 0 function, $0 \in U$. Let $f, g \in U$ be differentiable functions. Then $f + g$ is differentiable, so $f + g \in U$. Let $a \in \mathbb{R}$. Then af is differentiable, so $af \in U$. Therefore, U is a subspace of $\mathbb{R}^{\mathbb{R}}$.

(d) Suppose U is a subspace of $\mathbb{R}^{(0,3)}$. Then it must contain the 0 function on $(0, 3)$. Then $0' = 0 = b$, which implies $b = 0$.

Conversely, suppose $b = 0$. Then $U = \{f \in \mathbb{R}^{(0,3)} \mid f'(2) = 0\}$. Let $f, g \in U$ be differentiable functions such that $f'(2) = g'(2) = 0$. Then

$$(f + g)'(2) = f'(2) + g'(2) = 0 + 0 = 0,$$

so $f + g \in U$. Let $a \in \mathbb{R}$. Then

$$(af)'(2) = af'(2) = a \cdot 0 = 0,$$

so $af \in U$. Therefore, U is a subspace of $\mathbb{R}^{(0,3)}$.

(e) Let U be the set of all sequences of complex numbers with limit 0. Then the zero sequence is in U. Let $(z_n), (w_n) \in U$ be sequences with limit 0. Then

$$\lim_{n\to\infty} (z_n + w_n) = \lim_{n\to\infty} z_n + \lim_{n\to\infty} w_n = 0 + 0 = 0,$$

so $(z_n) + (w_n) \in U$. Let $a \in \mathbb{C}$. Then

$$\lim_{n\to\infty} (az_n) = a \lim_{n\to\infty} z_n = a \cdot 0 = 0,$$

so $a(z_n) \in U$. Therefore, U is a subspace of $\mathbb{C}^\infty$. ■

Exercise 1C3

Show that the set of differentiable real-valued functions f on the interval $(-4, 4)$ such that $f'(-1) = 3f(2)$ is a subspace of $\mathbb{R}^{(-4,4)}$.

Solution. Let U be the set in question. Then the zero function is in U since $0'(-1) = 0 = 3 \cdot 0 = 3 \cdot 0(2)$. Let $f, g \in U$ be differentiable functions such that $f'(-1) = 3f(2)$ and $g'(-1) = 3g(2)$. Then

$$(f + g)'(-1) = f'(-1) + g'(-1) = 3f(2) + 3g(2) = 3(f(2) + g(2)) = 3(f + g)(2),$$

so $f + g \in U$. Let $a \in \mathbb{R}$. Then

$$(af)'(-1) = af'(-1) = a \cdot 3f(2) = 3(af)(2),$$

so $af \in U$. Therefore, U is a subspace of $\mathbb{R}^{(-4,4)}$. ■

Exercise 1C4

Suppose $b \in \mathbb{R}$. Show that the set of continuous real-valued functions f on the interval $[0,1]$ such that

$$\int_0^1 f = b$$

is a subspace of $\mathbb{R}^{[0,1]}$ if and only if $b = 0$.

Solution. Let U be the set in question. Suppose U is a subspace of $\mathbb{R}^{[0,1]}$. Then it must contain the zero function on $[0,1]$. Then

$$\int_0^1 0 = 0 = b,$$

which implies $b = 0$.

Conversely, suppose $b = 0$. Then

$$U = \left\{ f \in \mathbb{R}^{[0,1]} \;\middle|\; \int_0^1 f = 0 \right\}.$$

The zero function is in U since $\int_0^1 0 = 0$. Let $f, g \in U$ be continuous functions such that $\int_0^1 f = \int_0^1 g = 0$. Then

$$\int_0^1 (f+g) = \int_0^1 f + \int_0^1 g = 0 + 0 = 0,$$

so $f + g \in U$. Let $a \in \mathbb{R}$. Then

$$\int_0^1 (af) = a \int_0^1 f = a \cdot 0 = 0,$$

so $af \in U$. Therefore, U is a subspace of $\mathbb{R}^{[0,1]}$. ■

Exercise 1C5

Is $\mathbb{R}^2$ a subspace of the complex vector space $\mathbb{C}^2$?

Solution. No, $\mathbb{R}^2$ is not a subspace of $\mathbb{C}^2$. Let $u = (1,0) \in \mathbb{R}^2$ and $a = i \in \mathbb{C}$. Then $au = (i,0) \notin \mathbb{R}^2$, so $\mathbb{R}^2$ is not closed under scalar multiplication and is not a subspace of $\mathbb{C}^2$. ■

Exercise 1C6

(a) Is $\{(a,b,c) \in \mathbb{R}^3 \mid a^3 = b^3\}$ a subspace of $\mathbb{R}^3$?

(b) Is $\{(a,b,c) \in \mathbb{C}^3 \mid a^3 = b^3\}$ a subspace of $\mathbb{C}^3$?

Remark. Here, I present an elementary argument that does not require the use of De Moivre's Theorem; rather, I arrive to the conclusion directly. However, readers who are familiar with it may find it easier to note that the complex number 1 has three cubic roots of unity. Other readers may also use Vieta's formulas to arrive at the same polynomial in ω to obtain $1 + \omega = -\omega^2$.

Solution. Let U be the subset in question.

(a) Since the zero vector $(0,0,0)$ satisfies $0^3 = 0^3$, U is nonempty. Let $x = (x_1, x_2, x_3)$ and $y = (y_1, y_2, y_3)$ be elements of U. Then $x_1^3 = x_2^3$ and $y_1^3 = y_2^3$ both imply that $x_1 = x_2$ and $y_1 = y_2$. Hence,

$$(x_1 + y_1)^3 = (x_2 + y_2)^3,$$

so $x+y \in U$. Let $a \in \mathbb{R}$. Then

$$(ax_1)^3 = a^3x_1^3 = a^3x_2^3 = (ax_2)^3,$$

so $ax \in U$. Therefore, U is a subspace of $\mathbb{R}^3$.

(b) No, U is not a subspace of $\mathbb{C}^3$. To see this, consider the cubic polynomial $x^3 - 1 = 0$. This polynomial has three distinct roots in $\mathbb{C}$ as follows:

$$x^3 - 1 = (x-1)(x^2+x+1) = 0$$

Applying the quadratic formula to $x^2+x+1=0$, we get

$$x = \frac{-1 \pm \sqrt{(-1)^2 - 4(1)(1)}}{2(1)} = \frac{-1 \pm i\sqrt{3}}{2}.$$

Let

$$\omega = \frac{-1+i\sqrt{3}}{2}.$$

Then $\omega^3 = 1$, and

$$\omega^2 = \frac{-1-i\sqrt{3}}{2}.$$

Moreover, it is not hard to see that $\omega + \omega^2 = -1$. Let $u = (1,1,0)$ and $v = (\omega, 1, 0)$. Then $u, v \in U$ since $1^3 = 1^3$ and $\omega^3 = 1^3$. Then $u+v = (1+\omega, 2, 0)$. However,

$$(1+\omega)^3 = 1 + 3\omega + 3\omega^2 + \omega^3 = 1 + 3(-1) + 1 = -1 \neq 2^3 = 8.$$

Hence, $u+v \notin U$, so U is not closed under addition and is not a subspace of $\mathbb{C}^3$. ■

Exercise 1C7

Prove or give a counterexample: If U is a nonempty subset of $\mathbb{R}^2$ such that U is closed under addition and under taking additive inverses (meaning $-u \in U$ whenever $u \in U$), then U is a subspace of $\mathbb{R}^2$.

Solution. Consider the subset $\mathbb{Q}^2$ of $\mathbb{R}^2$. Then $\mathbb{Q}^2$ is nonempty because $(0,0) \in \mathbb{Q}^2$, closed under addition since the sum of any two rational numbers is rational, and closed under taking additive inverses since the additive inverse of a rational number is rational. However, for any irrational number $r \in \mathbb{R}$, the vector $(r,0) \in \mathbb{R}^2$ is not in $\mathbb{Q}^2$. Hence, $\mathbb{Q}^2$ is not closed under scalar multiplication and is not a subspace of $\mathbb{R}^2$. Therefore, the statement is false. ■

Exercise 1C8

Give an example of a nonempty subset U of $\mathbb{R}^2$ such that U is closed under scalar multiplication, but U is not a subspace of $\mathbb{R}^2$.

Solution. Consider the subset

$$U = \{(x,0) \mid x \in \mathbb{R}\} \cup \{(0,y) \mid y \in \mathbb{R}\}.$$

Then U is nonempty since $(1,0) \in U$, and it is closed under scalar multiplication since any scalar multiple of $(1,0)$ or $(0,1)$ is still in U. However, U is not closed under addition since $(1,0)+(0,1) = (1,1) \notin U$. Hence, U is not a subspace of $\mathbb{R}^2$. ■

Exercise 1C9

A function $f : \mathbb{R} \to \mathbb{R}$ is called *periodic* if there exists a positive number p such that $f(x+p) = f(x)$ for all $x \in \mathbb{R}$. Is the set of periodic functions from $\mathbb{R}$ to $\mathbb{R}$ a subspace of $\mathbb{R}^{\mathbb{R}}$? Explain.

Solution. It is not a subspace of $\mathbb{R}^{\mathbb{R}}$. Consider the periodic functions $f(x) = \sin(2x)$ and $g(x) = \sin(\pi x)$. The function f has period π, and the function g has period 2. For $f(x+p) + g(x+p) = f(x) + g(x)$ to hold for all $x \in \mathbb{R}$, we would need a common period p that is a multiple of both π and 2. However, no such positive number exists since π is irrational. Therefore, the sum $f + g$ is not periodic, and the set of periodic functions is not closed under addition. Hence, it is not a subspace of $\mathbb{R}^{\mathbb{R}}$. ■

Exercise 1C10

Suppose V_1 and V_2 are subspaces of V. Prove that the intersection $V_1 \cap V_2$ is a subspace of V.

Solution. Let $U = V_1 \cap V_2$. Since both V_1 and V_2 are subspaces of V, they both contain the zero vector of V. Hence, $0 \in U$, so U is nonempty. Let $u, v \in U$. Then $u, v \in V_1$ and $u, v \in V_2$. Since both V_1 and V_2 are subspaces, they are closed under addition. Therefore, $u + v \in V_1$ and $u + v \in V_2$, which implies that $u + v \in U$. Let $a \in \mathbb{F}$. Then since both V_1 and V_2 are closed under scalar multiplication, we have that $au \in V_1$ and $au \in V_2$, which implies that $au \in U$. Therefore, U is a subspace of V. ■

Exercise 1C11

Prove that the intersection of every collection of subspaces of V is a subspace of V.

Solution. Let $\mathcal{C}$ be a collection of subspaces of V, and let

$$U = \bigcap_{W \in \mathcal{C}} W.$$

Since each subspace in $\mathcal{C}$ contains the zero vector, $0 \in U$, so U is nonempty. Let $u, v \in U$. Then $u, v \in W$ for all $W \in \mathcal{C}$. Since each W is a subspace, they are closed under addition, so $u + v \in W$ for all $W \in \mathcal{C}$, which implies $u + v \in U$. Let $a \in \mathbb{F}$. Then $au \in W$ for all $W \in \mathcal{C}$, which implies $au \in U$. Therefore, U is a subspace of V. ■

Exercise 1C12

Prove that the union of two subspaces of V is a subspace of V if and only if one of the subspaces is contained in the other.

Solution. Let U, W be subspaces of V. Suppose $U \subseteq W$. Then $U \cup W = W$, which is a subspace of V. Similarly, if $W \subseteq U$, then $U \cup W = U$, which is a subspace of V.

Conversely, suppose $U \cup W$ is a subspace of V. If neither $U \subseteq W$ nor $W \subseteq U$, then there exist $u \in U \setminus W$ and $w \in W \setminus U$. Since $U \cup W$ is a subspace, it must be closed under addition. Then $u + w \in U \cup W$. However, $u + w \notin U$ because if it were, then $w = (u + w) - u \in U$, which is a contradiction. Similarly, $u + w \notin W$ because if it were, then $u = (u + w) - w \in W$, which is also a contradiction. Hence, $u + w \notin U \cup W$, which contradicts the assumption that $U \cup W$ is a subspace. Therefore, one of the subspaces must be contained in the other. ■

Exercise 1C13

Prove that the union of three subspaces of V is a subspace of V if and only if one of the subspaces contains the other two.

Solution. Let U_1, U_2, U_3 be subspaces of V. Without loss of generality, suppose $U_2 \subseteq U_1$ and $U_3 \subseteq U_1$. Then $U_1 \cup U_2 \cup U_3 = U_1$, which is a subspace of V.

Conversely, suppose $U_1 \cup U_2 \cup U_3$ is a subspace of V. Note that $U_1 \cup U_2 \cup U_3 = (U_1 \cup U_2) \cup U_3$. Using Exercise 1C12, one of the subspaces $U_1 \cup U_2$ or U_3 must be contained in the other. If $U_1 \cup U_2 \subseteq U_3$, then $U_1 \cup U_2 \cup U_3 = U_3$, which is a subspace of V. If $U_3 \subseteq U_1 \cup U_2$, then $U_1 \cup U_2$ must be a subspace of V. By Exercise 1C12 again, then either $U_1 \subseteq U_2$ or $U_2 \subseteq U_1$. If the former is true, then $U_1 \cup U_2 = U_2$, hence U_2 contains U_1 and U_3. If the latter is true, then $U_1 \cup U_2 = U_1$, hence U_1 contains U_2 and U_3. Therefore, one of the subspaces must contain the other two. ■

Exercise 1C14

Suppose

$$U = \{(x, -x, 2x) \in \mathbb{F}^3 \mid x \in \mathbb{F}\} \quad \text{and} \quad W = \{(x, x, 2x) \in \mathbb{F}^3 \mid x \in \mathbb{F}\}.$$

Describe $U + W$ using symbols, and also give a description of $U + W$ that uses no symbols.

Solution. Let $u = (x, -x, 2x) \in U$ and $w = (y, y, 2y) \in W$. Then

$$u + w = (x + y, -x + y, 2x + 2y) = (x + y, y - x, 2(x + y)).$$

Observe that $x + y$ can take any value in $\mathbb{F}$, and $y - x$ can also take any value in $\mathbb{F}$. Hence, we can write

$$U + W = \{(a, b, 2a) \in \mathbb{F}^3 \mid a, b \in \mathbb{F}\}.$$

In words, $U + W$ is the set of all vectors in $\mathbb{F}^3$ whose third component is twice the first component, and whose second component can be any value in $\mathbb{F}$. ■

Exercise 1C15

Suppose U is a subspace of V. What is $U + U$?

Solution. Let $u_1, u_2 \in U$. Then $u_1 + u_2 \in U$ since U is closed under addition. Hence, $U + U \subseteq U$. Conversely, let $u \in U$. Then $u = u + 0$, where 0 is the additive identity in U. Since $0 \in U$, we have that $u \in U + U$. Hence, $U \subseteq U + U$. Therefore, $U + U = U$. ■

Exercise 1C16

Is the operation of addition on subspaces of V commutative? In other words, if U and W are subspaces of V, is $U + W = W + U$?

Solution. Yes, the operation of addition on subspaces of V is commutative. Let $u \in U$ and $w \in W$. Then $u + w \in U + W$. Because addition in V is commutative, we have that $u + w = w + u$, where $w \in W$ and $u \in U$. Hence, $w + u \in W + U$. Therefore, every element of $U + W$ is also an element of $W + U$, which implies that $U + W \subseteq W + U$. By a similar argument, we can show that $W + U \subseteq U + W$. Hence, $U + W = W + U$. ■

Exercise 1C17

Is the operation of addition on the subspaces of V associative? In other words, if V_1, V_2, V_3 are subspaces of V, is

$$(V_1 + V_2) + V_3 = V_1 + (V_2 + V_3)?$$

Solution. Yes, the operation of addition on subspaces of V is associative. Let $u_1 \in V_1$, $u_2 \in V_2$, and $u_3 \in V_3$. Then $(u_1 + u_2) + u_3 \in (V_1 + V_2) + V_3$. Because addition in V is associative, we have that $(u_1 + u_2) + u_3 = u_1 + (u_2 + u_3)$, where $u_1 \in V_1$, $u_2 \in V_2$, and $u_3 \in V_3$. Hence, $u_1 + (u_2 + u_3) \in V_1 + (V_2 + V_3)$. Therefore, every element of $(V_1+V_2)+V_3$ is also an element of $V_1+(V_2+V_3)$, which implies that $(V_1+V_2)+V_3 \subseteq V_1+(V_2+V_3)$. By a similar argument, we can show that $V_1+(V_2+V_3) \subseteq (V_1+V_2)+V_3$. Hence, $(V_1+V_2)+V_3 = V_1+(V_2+V_3)$. ■

Exercise 1C18

Does the operation of addition on the subspaces of V have an additive identity? Which subspaces have additive inverses?

Solution. Yes, the operation of addition on the subspaces of V has an additive identity. The additive identity is the zero subspace $\{0\}$, which contains only the zero vector of V. For any subspace U of V, we have that $U + \{0\} = U$.

If a subspace U of V had an additive inverse W, then we would have that $U + W = \{0\}$. However, $U \subseteq U + W$ since $u = u + 0$ for any $u \in U$. Hence, $U + W = \{0\}$ implies that $U = \{0\}$. Therefore, the only subspace of V that has an additive inverse is the zero subspace $\{0\}$. ■

Exercise 1C19

Prove or give a counterexample: If V_1, V_2, U are subspaces of V such that

$$V_1 + U = V_2 + U,$$

then $V_1 = V_2$.

Solution. The statement is false. Consider the vector space $\mathbb{R}^2$ and let

$$V_1 = \{(x, 0) \in \mathbb{R}^2 \mid x \in \mathbb{R}\}, \quad V_2 = \{(0, y) \in \mathbb{R}^2 \mid y \in \mathbb{R}\}, \quad U = \{(x, x) \in \mathbb{R}^2 \mid x \in \mathbb{R}\}.$$

Then $V_1 + U = \mathbb{R}^2$ and $V_2 + U = \mathbb{R}^2$, but $V_1 \neq V_2$. Therefore, the statement is false. ■

Exercise 1C20

Suppose

$$U = \{(x, x, y, y) \in \mathbb{F}^4 \mid x, y \in \mathbb{F}\}.$$

Find a subspace W of $\mathbb{F}^4$ such that $\mathbb{F}^4 = U \oplus W$.

Solution. Let

$$W = \{(0, z, 0, w) \in \mathbb{F}^4 \mid z, w \in \mathbb{F}\}.$$

If $(x, x, y, y) \in U$ and $(0, z, 0, w) \in W$, then $(x, x, y, y) = (0, z, 0, w)$ if and only if $x = 0$, $y = 0$, $z = 0$, and $w = 0$. Hence, $U \cap W = \{0\}$. Moreover, for any $(a, b, c, d) \in \mathbb{F}^4$, we can write

$$(a, b, c, d) = (x, x, y, y) + (0, z, 0, w),$$

where

$$x = a, \quad y = c, \quad z = b - x = b - a, \quad w = d - y = d - c.$$

Hence, every vector in $\mathbb{F}^4$ can be written as a sum of a vector in U and a vector in W. Therefore, $\mathbb{F}^4 = U \oplus W$. ■

Exercise 1C21

Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbb{F}^5 \mid x, y \in \mathbb{F}\}.$$

Find a subspace W of $\mathbb{F}^5$ such that $\mathbb{F}^5 = U \oplus W$.

Solution. Let

$$W = \{(0, 0, z, w, v) \in \mathbb{F}^5 \mid z, w, v \in \mathbb{F}\}.$$

If $(x, y, x + y, x - y, 2x) \in U$ and $(0, 0, z, w, v) \in W$, then $(x, y, x + y, x - y, 2x) = (0, 0, z, w, v)$ if and only if $x = 0$, $y = 0$, $z = 0$, $w = 0$, and $v = 0$. Hence, $U \cap W = \{0\}$. Moreover, for any $(a, b, c, d, e) \in \mathbb{F}^5$, we can write

$$(a, b, c, d, e) = (x, y, x + y, x - y, 2x) + (0, 0, z, w, v),$$

which implies

$$\begin{cases} a = x + 0 \\ b = y + 0 \\ c = (x + y) + z \\ d = (x - y) + w \\ e = 2x + v \end{cases} \implies \begin{cases} x = a \\ y = b \\ z = c - (x + y) = c - (a + b) \\ w = d - (x - y) = d - (a - b) \\ v = e - 2x = e - 2a \end{cases}$$

Hence, every vector in $\mathbb{F}^5$ can be written as a sum of a vector in U and a vector in W. Therefore, $\mathbb{F}^5 = U \oplus W$. ■

Exercise 1C22

Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbb{F}^5 \mid x, y \in \mathbb{F}\}.$$

Find three subspaces W_1, W_2, W_3 of $\mathbb{F}^5$, none of which equals $\{0\}$, such that $\mathbb{F}^5 = U \oplus W_1 \oplus W_2 \oplus W_3$.

Solution. Let

$$\begin{aligned} W_1 &= \{(0, 0, z, 0, 0) \in \mathbb{F}^5 \mid z \in \mathbb{F}\}, \\ W_2 &= \{(0, 0, 0, w, 0) \in \mathbb{F}^5 \mid w \in \mathbb{F}\}, \\ W_3 &= \{(0, 0, 0, 0, v) \in \mathbb{F}^5 \mid v \in \mathbb{F}\}. \end{aligned}$$

The intersection of any two of these subspaces is $\{0\}$, since they have no nonzero vectors in common. Additionally, the intersection of W_1, W_2, W_3 with U is also $\{0\}$, since any nonzero vector in U has nonzero entries in the first two coordinates, while any vector in W_1, W_2, or W_3 has zeros in the first two coordinates. The proof of showing that $\mathbb{F}^5 = U \oplus W_1 \oplus W_2 \oplus W_3$ is similar to that of Exercise 1C21, hence omitted. Therefore, $\mathbb{F}^5 = U \oplus W_1 \oplus W_2 \oplus W_3$. ■

Exercise 1C23

Prove or give a counterexample: If V_1, V_2, U are subspaces of V such that

$$V = V_1 \oplus U \quad \text{and} \quad V = V_2 \oplus U,$$

then $V_1 = V_2$.

Solution. See Exercise 1C19 for a counterexample. For $V_1 \oplus U$, observe that $V_1 \cap U = \{0\}$. Moreover, for any $(a, b) \in \mathbb{R}^2$, we can write

$$(a, b) = (a - b, 0) + (b, b) \in V_1 \oplus U.$$

The proof for $V_2 \oplus U$ is similar. Hence, $V = V_1 \oplus U = V_2 \oplus U$, but $V_1 \neq V_2$. Therefore, the statement is false. ■

Exercise 1C24

A function $f : \mathbb{R} \to \mathbb{R}$ is called *even* if

$$f(-x) = f(x)$$

for all $x \in \mathbb{R}$. A function $f : \mathbb{R} \to \mathbb{R}$ is called *odd* if

$$f(-x) = -f(x)$$

for all $x \in \mathbb{R}$. Let V_e denote the set of real-valued even functions on $\mathbb{R}$, and let V_o denote the set of real-valued odd functions on $\mathbb{R}$. Show that $\mathbb{R}^{\mathbb{R}} = V_e \oplus V_o$.

Solution. Let $f \in \mathbb{R}^{\mathbb{R}}$. Define

$$f_e(x) = \frac{f(x) + f(-x)}{2} \quad \text{and} \quad f_o(x) = \frac{f(x) - f(-x)}{2}.$$

Then f_e is even since

$$f_e(-x) = \frac{f(-x) + f(x)}{2} = f_e(x),$$

and f_o is odd since

$$f_o(-x) = \frac{f(-x) - f(x)}{2} = -\frac{f(x) - f(-x)}{2} = -f_o(x).$$

Moreover, we have that

$$f(x) = f_e(x) + f_o(x).$$

Hence, every function in $\mathbb{R}^{\mathbb{R}}$ can be written as a sum of an even function and an odd function.

Now, suppose $g \in V_e \cap V_o$. Then g is both even and odd, which implies that

$$g(0) = g(-0) = -g(0),$$

so $g(0) = 0$. For any $x \in \mathbb{R}$, we have that

$$g(x) = g(-(-x)) = -g(-x) = -g(x),$$

which implies that $g(x) = 0$. Hence, the only function in the intersection of V_e and V_o is the zero function. Therefore, $\mathbb{R}^{\mathbb{R}} = V_e \oplus V_o$. ■

2 Finite-Dimensional Vector Spaces

2A Span and Linear Independence

Exercise 2A1

Find a list of four distinct vectors in $\mathbb{F}^3$ whose span equals

$$\{(x, y, z) \in \mathbb{F}^3 \mid x + y + z = 0\}$$

Solution. Clearly, the span of the list involves two vectors, since we may rewrite z in terms of x and y as $z = -x - y$. Thus, we may choose the following two vectors:

$$v_1 = (1, 0, -1), \quad v_2 = (0, 1, -1).$$

To find two more vectors, we may pick any two distinct linear combinations of v_1 and v_2. For example, we may select

$$v_3 = v_1 + v_2 = (1, 1, -2), \quad v_4 = 2v_1 + 3v_2 = (2, 3, -5).$$

Then

$$\operatorname{span}(v_1, v_2, v_3, v_4) = \operatorname{span}(v_1, v_2) = \{(x, y, z) \in \mathbb{F}^3 \mid x + y + z = 0\}.$$

■

Exercise 2A2

Prove or give a counterexample: If v_1, v_2, v_3, v_4 spans V, then the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

also spans V.

Solution. Let $U = \{v_1, v_2, v_3, v_4\}$ and $W = \{v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4\}$. Clearly, $v_4 \in \operatorname{span} U$. Then we may write

$$\begin{aligned} v_3 &= (v_3 - v_4) + v_4 \in \operatorname{span} W, \\ v_2 &= (v_2 - v_3) + v_3 \in \operatorname{span} W, \\ v_1 &= (v_1 - v_2) + v_2 \in \operatorname{span} W. \end{aligned}$$

Thus, $v_1, v_2, v_3, v_4 \in \operatorname{span} W$, and so $V = \operatorname{span} U \subseteq \operatorname{span} W$. Since W is a list of vectors in V, we have $\operatorname{span} W \subseteq V$. Therefore, $\operatorname{span} W = V$, and so W spans V. ■

Exercise 2A3

Suppose $v_1, \ldots, v_m$ is a list of vectors in V. For $k \in \{1, \ldots, m\}$, let

$$w_k = v_1 + \cdots + v_k.$$

Show that $\operatorname{span}(v_1, \ldots, v_m) = \operatorname{span}(w_1, \ldots, w_m)$.

Solution. This is a generalization of Exercise 2A2 and a direct consequence of the fact that $v_k = w_k - w_{k-1}$ for $k \in \{2, \ldots, m\}$ and $v_1 = w_1$. Since each v_k is a linear combination of $w_1, \ldots, w_k$, we have $\operatorname{span}(v_1, \ldots, v_m) \subseteq \operatorname{span}(w_1, \ldots, w_m)$. On the other hand, each w_k is a linear combination of $v_1, \ldots, v_k$, so $\operatorname{span}(w_1, \ldots, w_m) \subseteq \operatorname{span}(v_1, \ldots, v_m)$. Therefore, $\operatorname{span}(v_1, \ldots, v_m) = \operatorname{span}(w_1, \ldots, w_m)$. ■

Exercise 2A4

(a) Show that a list of length one in a vector space is linearly independent if and only if the vector in the list is not 0.

(b) Show that a list of length two in a vector space is linearly independent if and only if neither of the two vectors in the list is a scalar multiple of the other.

Solution.

(a) Let v be the vector in the list. If $v = 0$, then the list is linearly dependent, since $0 \cdot v = 0$. If $v \neq 0$, then the only way to write 0 as a linear combination of v is to take the scalar multiple to be 0. Thus, the list is linearly independent.

(b) Let v_1 and v_2 be the vectors in the list. If v_1 is a scalar multiple of v_2, then there exists a scalar λ such that $v_1 = \lambda v_2$. Then we may write
$$0 = v_1 - \lambda v_2,$$
which shows that the list is linearly dependent. Conversely, if the list is linearly dependent, then there exist scalars α and β, not both zero, such that
$$\alpha v_1 + \beta v_2 = 0.$$
If $\alpha \neq 0$, then
$$v_1 = -\frac{\beta}{\alpha} v_2.$$
Similarly, if $\beta \neq 0$, then
$$v_2 = -\frac{\alpha}{\beta} v_1.$$
Thus, the list is linearly independent if and only if neither vector is a scalar multiple of the other. ■

Exercise 2A5

Find a number t such that
$$(3, 1, 4), (2, -3, 5), (5, 9, t)$$
is not linearly independent in $\mathbb{R}^3$.

Solution. To say that the list is not linearly independent is to say that there exist scalars α, β, γ, not all zero, such that
$$\alpha(3, 1, 4) + \beta(2, -3, 5) + \gamma(5, 9, t) = (0, 0, 0).$$
This is equivalent to the following system of equations:
$$\begin{aligned} 0 &= 3\alpha + 2\beta + 5\gamma, \\ 0 &= \alpha - 3\beta + 9\gamma, \\ 0 &= 4\alpha + 5\beta + t\gamma. \end{aligned}$$
From the second equation, we obtain $\alpha = 3\beta - 9\gamma$. Substituting this into the first equation, we have
$$\begin{aligned} 0 &= 3(3\beta - 9\gamma) + 2\beta + 5\gamma \\ &= 11\beta - 22\gamma. \end{aligned}$$

Thus, $\beta = 2\gamma$. Substituting this into the expression for α, we have $\alpha = 3(2\gamma) - 9\gamma = -3\gamma$. Substituting these values into the third equation, we have

$$\begin{aligned} 0 &= 4(-3\gamma) + 5(2\gamma) + t\gamma \\ &= -12\gamma + 10\gamma + t\gamma \\ &= (t-2)\gamma. \end{aligned}$$

Since $\gamma \neq 0$, we must have $t - 2 = 0$, or $t = 2$. Then the list is not linearly independent when $t = 2$. ■

Exercise 2A6

Show that the list $(2, 3, 1), (1, -1, 2), (7, 3, c)$ is linearly dependent in $\mathbb{F}^3$ if and only if $c = 8$.

Solution. We may proceed similarly to Exercise 2A5 to obtain the following system of equations:

$$\begin{aligned} 0 &= 2\alpha + \beta + 7\gamma, \\ 0 &= 3\alpha - \beta + 3\gamma, \\ 0 &= \alpha + 2\beta + c\gamma. \end{aligned}$$

The second equation gives us $\beta = 3\alpha + 3\gamma$. Substituting this into the first equation, we have

$$\begin{aligned} 0 &= 2\alpha + (3\alpha + 3\gamma) + 7\gamma \\ &= 5\alpha + 10\gamma. \end{aligned}$$

Thus, $\alpha = -2\gamma$. Substituting this into the expression for β, we have $\beta = 3(-2\gamma) + 3\gamma = -3\gamma$. Substituting these values into the third equation, we have

$$\begin{aligned} 0 &= (-2\gamma) + 2(-3\gamma) + c\gamma \\ &= -2\gamma - 6\gamma + c\gamma \\ &= (c-8)\gamma. \end{aligned}$$

Since $\gamma \neq 0$, we must have $c - 8 = 0$, or $c = 8$. Then the list is linearly dependent when $c = 8$.

Conversely, if $c \neq 8$, then $(c-8)\gamma = 0$ implies $\gamma = 0$. Then $\alpha = -2\gamma = 0$ and $\beta = -3\gamma = 0$, so the only solution to the system of equations is $\alpha = \beta = \gamma = 0$. Thus, the list is linearly independent when $c \neq 8$. ■

Exercise 2A7

(a) Show that if we think of $\mathbb{C}$ as a vector space over $\mathbb{R}$, then the list $1+i, 1-i$ is linearly independent.

(b) Show that if we think of $\mathbb{C}$ as a vector space over $\mathbb{C}$, then the list $1+i, 1-i$ is linearly dependent.

Solution.

(a) Suppose $\alpha, \beta \in \mathbb{R}$ such that

$$\alpha(1+i) + \beta(1-i) = 0.$$

Then we have

$$(\alpha + \beta) + (\alpha - \beta)i = 0.$$

Since $0 = 0 + 0i$, we must have $\alpha + \beta = 0$ and $\alpha - \beta = 0$. Solving this system of equations, we find that $\alpha = \beta = 0$. Thus, the list is linearly independent.

(b) Suppose $a + bi, c + di \in \mathbb{C}$ such that

$$(a + bi)(1 + i) + (c + di)(1 - i) = 0.$$

Then we have

$$(a - b + c + d) + (a + b - c + d)i = 0.$$

Then we have the system of equations

$$\begin{aligned} 0 &= a - b + c + d, \\ 0 &= a + b - c + d. \end{aligned}$$

Adding the two equations results in $0 = 2a + 2d$, or $a = -d$. Substituting this into the first equation, we obtain $b = c$. Then we have infinitely many nontrivial solutions to the system of equations, such as $a = 1, b = 0, c = 0, d = -1$. Thus, the list is linearly dependent. ∎

Exercise 2A8

Suppose v_1, v_2, v_3, v_4 is linearly independent in V. Prove that the list $v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$ is also linearly independent.

Solution. Let $\alpha, \beta, \gamma, \delta \in \mathbb{F}$ such that

$$\alpha(v_1 - v_2) + \beta(v_2 - v_3) + \gamma(v_3 - v_4) + \delta v_4 = 0.$$

Then we have

$$\alpha v_1 + (-\alpha + \beta)v_2 + (-\beta + \gamma)v_3 + (-\gamma + \delta)v_4 = 0.$$

Since v_1, v_2, v_3, v_4 is linearly independent, we must have

$$\begin{aligned} 0 &= \alpha, \\ 0 &= -\alpha + \beta, \\ 0 &= -\beta + \gamma, \\ 0 &= -\gamma + \delta. \end{aligned}$$

From the first equation, we have $\alpha = 0$. Substituting this into the second equation gives us $\beta = 0$. Substituting this into the third equation gives us $\gamma = 0$. Finally, substituting this into the fourth equation gives us $\delta = 0$. Thus, the only solution to the original equation is $\alpha = \beta = \gamma = \delta = 0$, and so the list is linearly independent. ∎

Exercise 2A9

Prove or give a counterexample: If $v_1, v_2, \ldots, v_m$ is a linearly independent list of vectors in V, then $5v_1 - 4v_2, v_2, v_3, \ldots, v_m$ is linearly independent.

Solution. Let $\alpha_1, \ldots, \alpha_m \in \mathbb{F}$ such that

$$\alpha_1(5v_1 - 4v_2) + \alpha_2 v_2 + \cdots + \alpha_m v_m = 0.$$

Then we have

$$5\alpha_1 v_1 + (-4\alpha_1 + \alpha_2)v_2 + \alpha_3 v_3 + \cdots + \alpha_m v_m = 0.$$

Since $v_1, v_2, \ldots, v_m$ is linearly independent, we must have

$$\begin{aligned} 0 &= 5\alpha_1, \\ 0 &= -4\alpha_1 + \alpha_2, \\ 0 &= \alpha_3, \\ &\vdots \\ 0 &= \alpha_m. \end{aligned}$$

From the first equation, we have $\alpha_1 = 0$. Substituting this into the second equation gives us $\alpha_2 = 0$. The remaining equations give us $\alpha_k = 0$ for $k = 3, \ldots, m$. Thus, the only solution to the original equation is $\alpha_k = 0$ for $k = 1, \ldots, m$, and so the list is linearly independent. ■

Exercise 2A10

Prove or give a counterexample: If $v_1, v_2, \ldots, v_m$ is a linearly independent list of vectors in V and $\lambda \in \mathbb{F}$ with $\lambda \neq 0$, then $\lambda v_1, \lambda v_2, \ldots, \lambda v_m$ is linearly independent.

Solution. Let $\alpha_1, \ldots, \alpha_m \in \mathbb{F}$ such that

$$\alpha_1(\lambda v_1) + \cdots + \alpha_m(\lambda v_m) = 0.$$

Then we have

$$\lambda(\alpha_1 v_1 + \cdots + \alpha_m v_m) = 0.$$

Since $\lambda \neq 0$, we must have

$$\alpha_1 v_1 + \cdots + \alpha_m v_m = 0.$$

Since $v_1, v_2, \ldots, v_m$ is linearly independent, we must have $\alpha_k = 0$ for $k = 1, \ldots, m$. Thus, the only solution to the original equation is $\alpha_k = 0$ for $k = 1, \ldots, m$, and so the list is linearly independent. ■

Exercise 2A11

Prove or give a counterexample: If $v_1, \ldots, v_m$ and $w_1, \ldots, w_m$ are linearly independent lists of vectors in V, then the list $v_1 + w_1, \ldots, v_m + w_m$ is linearly independent.

Solution. Consider the following counterexample in $\mathbb{R}^2$:

$$v_1 = (1, 0), \quad v_2 = (0, 1), \quad w_1 = (0, 1), \quad w_2 = (1, 0).$$

Then v_1, v_2 and w_1, w_2 are linearly independent lists of vectors in $\mathbb{R}^2$. However,

$$v_1 + w_1 = (1, 1), \quad v_2 + w_2 = (1, 1),$$

and so the list $v_1 + w_1, v_2 + w_2$ is linearly dependent. Thus, the statement is false. ■

Exercise 2A12

Suppose $v_1, \ldots, v_m$ is linearly independent in V and $w \in V$. Prove that if $v_1 + w, \ldots, v_m + w$ is linearly dependent, then $w \in \operatorname{span}(v_1, \ldots, v_m)$.

Solution. Let $\alpha_1, \ldots, \alpha_m \in \mathbb{F}$, not all zero, such that

$$\alpha_1(v_1 + w) + \cdots + \alpha_m(v_m + w) = 0.$$

Then we have

$$(\alpha_1 v_1 + \cdots + \alpha_m v_m) + (\alpha_1 + \cdots + \alpha_m)w = 0.$$

If $\alpha_1 + \cdots + \alpha_m = 0$, then we have

$$\alpha_1 v_1 + \cdots + \alpha_m v_m = 0,$$

which contradicts the linear independence of $v_1, \ldots, v_m$. Thus, we must have $\alpha_1 + \cdots + \alpha_m \neq 0$. Let $\beta = \alpha_1 + \cdots + \alpha_m$. Then we may write

$$w = -\frac{\alpha_1}{\beta} v_1 - \cdots - \frac{\alpha_m}{\beta} v_m.$$

Thus, w is a linear combination of $v_1, \ldots, v_m$, and so $w \in \operatorname{span}(v_1, \ldots, v_m)$. ■

Exercise 2A13

Suppose $v_1, \ldots, v_m$ is linearly independent in V and $w \in V$. Show that

$$v_1, \ldots, v_m, w \text{ is linearly independent} \iff w \notin \operatorname{span}(v_1, \ldots, v_m).$$

Solution. Suppose $v_1, \ldots, v_m, w$ is linearly independent. If there were scalars $\alpha_1, \ldots, \alpha_m, \beta$ such that

$$\alpha_1 v_1 + \cdots + \alpha_m v_m + \beta w = 0,$$

then linear independence would imply that $\alpha_1 = \cdots = \alpha_m = \beta = 0$. Suppose $w \in \operatorname{span}(v_1, \ldots, v_m)$. Then there exist scalars $\gamma_1, \ldots, \gamma_m$ not all zero such that

$$\gamma_1 v_1 + \cdots + \gamma_m v_m = w.$$

Then we may write

$$-\gamma_1 v_1 - \cdots - \gamma_m v_m + 1 \cdot w = 0,$$

which contradicts the linear independence of $v_1, \ldots, v_m, w$. Thus, $w \notin \operatorname{span}(v_1, \ldots, v_m)$.

Conversely, suppose $w \notin \operatorname{span}(v_1, \ldots, v_m)$. Then there do not exist scalars $\alpha_1, \ldots, \alpha_m$ such that

$$\alpha_1 v_1 + \cdots + \alpha_m v_m = w.$$

If $v_1, \ldots, v_m, w$ is linearly dependent, then there exist scalars $\beta_1, \ldots, \beta_m, \gamma$, not all zero, such that

$$\beta_1 v_1 + \cdots + \beta_m v_m + \gamma w = 0.$$

If $\gamma = 0$, then we have

$$\beta_1 v_1 + \cdots + \beta_m v_m = 0,$$

which contradicts the linear independence of $v_1, \ldots, v_m$. Thus, we must have $\gamma \neq 0$. Then we may write

$$w = -\frac{\beta_1}{\gamma} v_1 - \cdots - \frac{\beta_m}{\gamma} v_m,$$

which contradicts the assumption that w is not in the span of $v_1, \ldots, v_m$. Thus, $v_1, \ldots, v_m, w$ is linearly independent. ■

Exercise 2A14

Suppose $v_1, \ldots, v_m$ is a list of vectors in V. For $k \in \{1, \ldots, m\}$, let

$$w_k = v_1 + \cdots + v_k.$$

Show that the list $v_1, \ldots, v_m$ is linearly independent if and only if the list $w_1, \ldots, w_m$ is linearly independent.

Solution. Suppose $v_1, \ldots, v_m$ is linearly independent. Let $\alpha_1, \ldots, \alpha_m \in \mathbb{F}$ such that

$$\alpha_1 w_1 + \cdots + \alpha_m w_m = 0.$$

Noting that $v_k = w_k - w_{k-1}$ for $k \in \{2, \ldots, m\}$ and $v_1 = w_1$, we may rewrite the equation as

$$\alpha_1 v_1 + (\alpha_2 - \alpha_1) v_2 + \cdots + (\alpha_m - \alpha_{m-1}) v_m = 0.$$

Since $v_1, \ldots, v_m$ is linearly independent, we must have $\alpha_1 = 0$. Then this implies that $\alpha_2 - \alpha_1 = 0$, and so on, until we have $\alpha_m - \alpha_{m-1} = 0$. Thus, the only solution to the original equation is $\alpha_1 = \cdots = \alpha_m = 0$, and so $w_1, \ldots, w_m$ is linearly independent.

Conversely, suppose $w_1, \ldots, w_m$ is linearly independent. Let $\beta_1, \ldots, \beta_m \in \mathbb{F}$ such that

$$\beta_1 v_1 + \cdots + \beta_m v_m = 0.$$

Then we have

$$\beta_1 w_1 + (\beta_2 + \beta_1) w_2 + \cdots + (\beta_m + \cdots + \beta_1) w_m = 0.$$

Since $w_1, \ldots, w_m$ is linearly independent, we must have $\beta_1 = 0$. Then this implies that $\beta_2 + \beta_1 = 0$, and so on, until we have $\beta_m + \cdots + \beta_1 = 0$. Thus, the only solution to the original equation is $\beta_1 = \cdots = \beta_m = 0$, and so $v_1, \ldots, v_m$ is linearly independent. ■

Exercise 2A15

Explain why there does not exist a list of six polynomials that is linearly independent in $\mathcal{P}_4(\mathbb{F})$.

Solution. Since $\mathcal{P}_4(\mathbb{F})$ can be spanned by the list $1, x, x^2, x^3, x^4$, and the length of a linearly independent list cannot exceed the length of a spanning list, there does not exist a list of six polynomials that is linearly independent in $\mathcal{P}_4(\mathbb{F})$. ■

Exercise 2A16

Explain why no list of four polynomials spans $\mathcal{P}_4(\mathbb{F})$.

Solution. Since $\mathcal{P}_4(\mathbb{F})$ can be spanned by the list $1, x, x^2, x^3, x^4$, and the length of a spanning list cannot be less than the length of a linearly independent list, no list of four polynomials spans $\mathcal{P}_4(\mathbb{F})$. ■

Exercise 2A17

Prove that V is infinite-dimensional if and only if there is a sequence $v_1, v_2, \ldots$ of vectors in V such that $v_1, \ldots, v_m$ is linearly independent for every positive integer m.

Solution. Suppose V is infinite-dimensional. We proceed by induction to construct a sequence of vectors $v_1, v_2, \ldots$ in V such that $v_1, \ldots, v_m$ is linearly independent for every positive integer m. Since V is infinite-dimensional, there exists a nonzero vector $v_1 \in V$. Suppose we have constructed a linearly independent list of vectors $v_1, \ldots, v_m$ in V. Since V is infinite-dimensional, there exists a vector $v_{m+1} \in V$ such that $v_{m+1} \notin \operatorname{span}(v_1, \ldots, v_m)$. Then $v_1, \ldots, v_m, v_{m+1}$ is linearly independent. By induction, we have constructed a sequence of vectors $v_1, v_2, \ldots$ in V such that $v_1, \ldots, v_m$ is linearly independent for every positive integer m.

Conversely, suppose there exists a sequence of vectors $v_1, v_2, \ldots$ in V such that $v_1, \ldots, v_m$ is linearly independent for every positive integer m. Then for every positive integer m, there exists a linearly independent list of vectors in V of length m. Since there is no maximum length for a linearly independent list of vectors in V, this implies that V is infinite-dimensional. ■

Exercise 2A18

Prove that $\mathbb{F}^\infty$ is infinite-dimensional.

Solution. By Exercise 2A17, it suffices to construct a sequence of vectors $v_1, v_2, \ldots$ in $\mathbb{F}^\infty$ such that $v_1, \ldots, v_m$ is linearly independent for every positive integer m. Let v_k be the vector in $\mathbb{F}^\infty$ whose k-th coordinate is 1 and all other coordinates are 0. Explicitly, we have

$$v_1 = (1, 0, 0, \ldots), \quad v_2 = (0, 1, 0, \ldots), \quad v_3 = (0, 0, 1, \ldots), \quad \ldots$$

Then for any positive integer m, the list $v_1, \ldots, v_m$ is linearly independent, since the only way to write the zero vector as a linear combination of $v_1, \ldots, v_m$ is to take all coefficients to be zero. Thus, by Exercise 2A17, $\mathbb{F}^\infty$ is infinite-dimensional. ∎

Exercise 2A19

Prove that the real vector space of all continuous real-valued functions on the interval $[0, 1]$ is infinite-dimensional.

Solution. By Exercise 2A17, it suffices to construct a sequence of continuous real-valued functions $f_1, f_2, \ldots$ on the interval $[0, 1]$ such that $f_1, \ldots, f_m$ is linearly independent for every positive integer m. Let $f_k(x) = x^{k-1}$ for $k = 1, 2, \ldots$. Then for any positive integer m, the list $f_1, \ldots, f_m$ is linearly independent, since the only way to write the zero function as a linear combination of $f_1, \ldots, f_m$ is to take all coefficients to be zero. Thus, by Exercise 2A17, $\mathbb{R}^{[0,1]}$ is infinite-dimensional. ∎

Exercise 2A20

Suppose $p_0, p_1, \ldots, p_m$ are polynomials in $\mathcal{P}_m(\mathbb{F})$ such that $p_k(2) = 0$ for each $k \in \{0, \ldots, m\}$. Prove that $p_0, p_1, \ldots, p_m$ is not linearly independent in $\mathcal{P}_m(\mathbb{F})$.

Solution. Since $p_k(2) = 0$ for each $k \in \{0, \ldots, m\}$, we have

$$p_k(x) = (x - 2)q_k(x)$$

for some polynomial $q_k(x) \in \mathcal{P}_{m-1}(\mathbb{F})$. Then we may write

$$p_0(x), p_1(x), \ldots, p_m(x) = (x - 2)q_0(x), (x - 2)q_1(x), \ldots, (x - 2)q_m(x).$$

Since $\mathcal{P}_{m-1}(\mathbb{F})$ can be spanned by the m polynomials $1, x, \ldots, x^{m-1}$, the list of $m+1$ polynomials $q_0, q_1, \ldots, q_m$ must be linearly dependent. Thus, there exist scalars $\alpha_0, \ldots, \alpha_m$, not all zero, such that

$$\alpha_0 q_0(x) + \cdots + \alpha_m q_m(x) = 0.$$

Then we may write

$$\alpha_0 p_0(x) + \cdots + \alpha_m p_m(x) = (x - 2)(\alpha_0 q_0(x) + \cdots + \alpha_m q_m(x)) = 0.$$

Since not all the scalars $\alpha_0, \ldots, \alpha_m$ are zero, this shows that $p_0, p_1, \ldots, p_m$ is linearly dependent in $\mathcal{P}_m(\mathbb{F})$. ∎

2B Bases

Exercise 2B1

Find all vector spaces that have exactly one basis.

Solution. Let V be a vector space. If V has exactly one basis, then V must be finite-dimensional, since infinite-dimensional vector spaces have infinitely many bases. Let $v_1, \dots, v_n$ be a basis of V. Then any other basis of V must also have length n. Since there is only one basis of V, it must be that $n = 0$, and so the only vector space that has exactly one basis is the zero vector space. ■

Exercise 2B2

Verify all assertions in Example 2.27.

(a) The list $(1, 0, \dots, 0), (0, 1, 0, \dots, 0), \dots, (0, \dots, 0, 1)$ is a basis of $\mathbb{F}^n$, called the *standard basis* of $\mathbb{F}^n$.

(b) The list $(1, 2), (3, 5)$ is a basis of $\mathbb{F}^2$. Note that this list has length two, which is the same as the length of the standard basis of $\mathbb{F}^2$. In the next section, we will see that this is not a coincidence.

(c) The list $(1, 2, -4), (7, -5, 6)$ is linearly independent in $\mathbb{F}^3$ but is not a basis of $\mathbb{F}^3$ because it does not span $\mathbb{F}^3$.

(d) The list $(1, 2), (3, 5), (4, 13)$ spans $\mathbb{F}^2$ but is not a basis of $\mathbb{F}^2$ because it is not linearly independent.

(e) The list $(1, 1, 0), (0, 0, 1)$ is a basis of $\{(x, x, y) \in \mathbb{F}^3 \mid x, y \in \mathbb{F}\}$.

(f) The list $(1, -1, 0), (1, 0, -1)$ is a basis of

$$\{(x, y, z) \in \mathbb{F}^3 \mid x + y + z = 0\}.$$

(g) The list $1, z, \dots, z^m$ is a basis of $\mathcal{P}_m(\mathbb{F})$, called the *standard basis* of $\mathcal{P}_m(\mathbb{F})$.

Solution.

(a) Suppose $\alpha_1, \dots, \alpha_n \in \mathbb{F}$ such that

$$\alpha_1(1, 0, \dots, 0) + \alpha_2(0, 1, 0, \dots, 0) + \cdots + \alpha_n(0, \dots, 0, 1) = 0.$$

Then each $\alpha_i = 0$, so the list is linearly independent. Moreover, any vector $(x_1, \dots, x_n) \in \mathbb{F}^n$ can be written as

$$(x_1, \dots, x_n) = x_1(1, 0, \dots, 0) + x_2(0, 1, 0, \dots, 0) + \cdots + x_n(0, \dots, 0, 1),$$

so the list spans $\mathbb{F}^n$. Thus, the list is a basis of $\mathbb{F}^n$.

(b) If $(3, 5) = a(1, 2)$ for some $a \in \mathbb{F}$, then we must have $3 = a$ and $5 = 2a$, which is impossible. Thus, the list is linearly independent. Moreover, for any element $(x, y) \in \mathbb{F}^2$, we can solve the system of equations

$$x = \alpha + 3\beta,$$
$$y = 2\alpha + 5\beta$$

for $\alpha, \beta \in \mathbb{F}$. Solving this system of equations, we find that

$$\alpha = 5x - 3y, \quad \beta = -2x + y.$$

Since we can find $\alpha, \beta \in \mathbb{F}$ for any $(x, y) \in \mathbb{F}^2$, the list spans $\mathbb{F}^2$. Thus, the list is a basis of $\mathbb{F}^2$.

(c) If $(7, -5, 6) = a(1, 2, -4)$ for some $a \in \mathbb{F}$, then we must have $7 = a$, $-5 = 2a$, and $6 = -4a$, which is impossible. Thus, the list is linearly independent. Moreover, the list does not span $\mathbb{F}^3$. If it did, then we could write $(0, 0, 1) = \alpha(1, 2, -4) + \beta(7, -5, 6)$ for some $\alpha, \beta \in \mathbb{F}$. This would give us the system of equations

$$\begin{aligned} 0 &= \alpha + 7\beta, \\ 0 &= 2\alpha - 5\beta, \\ 1 &= -4\alpha + 6\beta. \end{aligned}$$

From the first equation, we see that $\alpha = -7\beta$. Substituting this into the second equation gives us $0 = -14\beta - 5\beta = -19\beta$, so $\beta = 0$ and $\alpha = 0$. However, substituting these values into the third equation gives us $1 = 0$, which is impossible. Thus, the list does not span $\mathbb{F}^3$, and so it is not a basis of $\mathbb{F}^3$.

(d) From part (b), we saw that $(1, 2), (3, 5)$ is a basis of $\mathbb{F}^2$. Since $(4, 13) = 5(1, 2) - 2(3, 5)$, the list $(1, 2), (3, 5), (4, 13)$ is linearly dependent. Moreover, since $(1, 2), (3, 5)$ is a basis of $\mathbb{F}^2$, the list $(1, 2), (3, 5), (4, 13)$ spans $\mathbb{F}^2$. Thus, the list is not a basis of $\mathbb{F}^2$.

(e) Let U be the subspace in question. If $\alpha, \beta \in \mathbb{F}$ such that

$$\alpha(1, 1, 0) + \beta(0, 0, 1) = (0, 0, 0),$$

then we must have $\alpha = \beta = 0$, so the list is linearly independent. Moreover, for any $(x, x, y) \in U$, we can write

$$(x, x, y) = x(1, 1, 0) + y(0, 0, 1),$$

so the list spans U. Thus, the list is a basis of U.

(f) Let U be the subspace in question. If $\alpha, \beta \in \mathbb{F}$ such that

$$\alpha(1, -1, 0) + \beta(1, 0, -1) = (0, 0, 0),$$

then we must have $\alpha = \beta = 0$, so the list is linearly independent. Moreover, for any $(x, y, z) \in U$, we can write

$$(x, y, z) = (x - z)(1, -1, 0) + (y - z)(1, 0, -1),$$

so the list spans U. Thus, the list is a basis of U.

(g) If $\alpha_0, \ldots, \alpha_m \in \mathbb{F}$ such that

$$\alpha_0 + \alpha_1 z + \cdots + \alpha_m z^m = 0,$$

then we must have $\alpha_0 = \cdots = \alpha_m = 0$, so the list is linearly independent. Moreover, any polynomial $p(z) \in \mathcal{P}_m(\mathbb{F})$ can be written as

$$p(z) = \alpha_0 + \alpha_1 z + \cdots + \alpha_m z^m,$$

so the list spans $\mathcal{P}_m(\mathbb{F})$. Thus, the list is a basis of $\mathcal{P}_m(\mathbb{F})$. ■

Exercise 2B3

(a) Let U be the subspace of $\mathbb{R}^5$ defined by

$$U = \{(x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5 \mid x_1 = 3x_2 \text{ and } x_3 = 7x_4\}.$$

Find a basis of U.

(b) Extend the basis in (a) to a basis of $\mathbb{R}^5$.

(c) Find a subspace W of $\mathbb{R}^5$ such that $\mathbb{R}^5 = U \oplus W$.

Solution.

(a) Note that we can rewrite the subspace U as

$$U = \{(3x_2, x_2, 7x_4, x_4, x_5) \in \mathbb{R}^5 \mid x_2, x_4, x_5 \in \mathbb{R}\}.$$

Then we can write any vector in U as a linear combination of the following three vectors:

$$u_1 = (3, 1, 0, 0, 0), \quad u_2 = (0, 0, 7, 1, 0), \quad u_3 = (0, 0, 0, 0, 1).$$

Moreover, these three vectors are linearly independent, since the only way to write the zero vector as a linear combination of u_1, u_2, u_3 is to take all coefficients to be zero. Thus, $\{u_1, u_2, u_3\}$ is a basis of U.

(b) We can extend the basis $\{u_1, u_2, u_3\}$ of U to a basis of $\mathbb{R}^5$ by adding two more vectors that are linearly independent of u_1, u_2, u_3. For example, we can add the vectors

$$u_4 = (1, 0, 0, 0, 0), \quad u_5 = (0, 0, 1, 0, 0),$$

which are linearly independent of u_1, u_2, u_3 since they are not in U. Thus, $\{u_1, u_2, u_3, u_4, u_5\}$ is a basis of $\mathbb{R}^5$.

(c) Let W be the subspace of $\mathbb{R}^5$ spanned by the vectors u_4 and u_5. Then we have

$$\mathbb{R}^5 = U \oplus W,$$

since $U \cap W = \{0\}$ and every vector in $\mathbb{R}^5$ can be written as a sum of a vector in U and a vector in W. ■

Exercise 2B4

(a) Let U be the subspace of $\mathbb{C}^5$ defined by

$$U = \{(z_1, z_2, z_3, z_4, z_5) \in \mathbb{C}^5 \mid 6z_1 = z_2 \text{ and } z_3 + 2z_4 + 3z_5 = 0\}.$$

Find a basis of U.

(b) Extend the basis in (a) to a basis of $\mathbb{C}^5$.

(c) Find a subspace W of $\mathbb{C}^5$ such that $\mathbb{C}^5 = U \oplus W$.

Solution.

(a) Note that we can rewrite the subspace U as

$$U = \{(z_1, 6z_1, -2z_4 - 3z_5, z_4, z_5) \in \mathbb{C}^5 \mid z_1, z_4, z_5 \in \mathbb{C}\}.$$

Then we can write any vector in U as a linear combination of the following three vectors:

$$u_1 = (1, 6, 0, 0, 0), \quad u_2 = (0, 0, -2, 1, 0), \quad u_3 = (0, 0, -3, 0, 1).$$

Moreover, these three vectors are linearly independent, since the only way to write the zero vector as a linear combination of u_1, u_2, u_3 is to take all coefficients to be zero. Thus, $\{u_1, u_2, u_3\}$ is a basis of U.

(b) We can extend the basis $\{u_1, u_2, u_3\}$ of U to a basis of $\mathbb{C}^5$ by adding two more vectors that are linearly independent of u_1, u_2, u_3. For example, we can add the vectors

$$u_4 = (1, 0, 0, 0, 0), \quad u_5 = (0, 0, 1, 0, 0),$$

which are linearly independent of u_1, u_2, u_3 since they are not in U. Thus, $\{u_1, u_2, u_3, u_4, u_5\}$ is a basis of $\mathbb{C}^5$.

(c) Let W be the subspace of $\mathbb{C}^5$ spanned by the vectors u_4 and u_5. Then we have

$$\mathbb{C}^5 = U \oplus W,$$

since $U \cap W = \{0\}$ and every vector in $\mathbb{C}^5$ can be written as a sum of a vector in U and a vector in W. ■

Exercise 2B5

Suppose V is finite-dimensional and U, W are subspaces of V such that $V = U + W$. Prove that there exists a basis of V consisting of vectors in $U \cup W$.

Solution. Let $u_1, \ldots, u_m$ be a basis of U and $w_1, \ldots, w_n$ be a basis of W. Then the list $u_1, \ldots, u_m, w_1, \ldots, w_n$ spans V, since any vector in V can be written as a sum of a vector in U and a vector in W. We can extract a basis of V from this list by removing any linearly dependent vectors. Thus, there exists a basis of V consisting of vectors in $U \cup W$. ■

Exercise 2B6

Prove or give a counterexample: If p_0, p_1, p_2, p_3 is a list in $\mathcal{P}_3(\mathbb{F})$ such that none of the polynomials p_0, p_1, p_2, p_3 has degree 2, then the list is not a basis of $\mathcal{P}_3(\mathbb{F})$.

Solution. This is not true. Let $p_0 = 1, p_1 = x, p_2 = x^3, p_3 = x^2 + x^3$. Then none of the polynomials p_0, p_1, p_2, p_3 has degree 2. Moreover, the list p_0, p_1, p_2, p_3 is linearly independent, since the only way to write the zero polynomial as a linear combination of p_0, p_1, p_2, p_3 is to take all coefficients to be zero. Finally, noting that $x^2 = p_3 - p_2$, then for any polynomial $p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 \in \mathcal{P}_3(\mathbb{F})$, we can write

$$p(x) = a_0 p_0 + a_1 p_1 + a_2(p_3 - p_2) + a_3 p_2 = a_0 p_0 + a_1 p_1 + (a_3 - a_2) p_2 + a_2 p_3,$$

which shows that the list spans $\mathcal{P}_3(\mathbb{F})$. Thus, the list p_0, p_1, p_2, p_3 is a basis of $\mathcal{P}_3(\mathbb{F})$. ■

Exercise 2B7

Suppose v_1, v_2, v_3, v_4 is a basis of V. Prove that

$$v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$$

is also a basis of V.

Solution. Let $\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathbb{F}$ such that

$$\alpha_1(v_1 + v_2) + \alpha_2(v_2 + v_3) + \alpha_3(v_3 + v_4) + \alpha_4 v_4 = 0.$$

Then we may rewrite this equation as

$$\alpha_1 v_1 + (\alpha_1 + \alpha_2) v_2 + (\alpha_2 + \alpha_3) v_3 + (\alpha_3 + \alpha_4) v_4 = 0.$$

Since v_1, v_2, v_3, v_4 is a basis of V, it is linearly independent, and so we must have

$$\alpha_1 = 0, \quad \alpha_1 + \alpha_2 = 0, \quad \alpha_2 + \alpha_3 = 0, \quad \alpha_3 + \alpha_4 = 0.$$

From the first equation, we have $\alpha_1 = 0$. Then the second equation gives us $\alpha_2 = 0$. The third equation then gives us $\alpha_3 = 0$, and finally the fourth equation gives us $\alpha_4 = 0$. Thus, the list is linearly independent. Moreover, since v_1, v_2, v_3, v_4 is a basis of V, it spans V, and so the list also spans V. Therefore, the list is a basis of V. ■

Exercise 2B8

Prove or give a counterexample: If v_1, v_2, v_3, v_4 is a basis of V and U is a subspace of V such that $v_1, v_2 \in U$ and $v_3 \notin U$ and $v_4 \notin U$, then v_1, v_2 is a basis of U.

Solution. This is not true. Let $V = \mathbb{R}^4$ and v_1, v_2, v_3, v_4 be the standard basis of $\mathbb{R}^4$. Let U be the subspace of $\mathbb{R}^4$ defined by

$$U = \operatorname{span}(v_1, v_2, v_3 + v_4) = \{(x, y, z, z) \in \mathbb{R}^4 \mid x, y, z \in \mathbb{R}\}.$$

Then $v_1, v_2 \in U$ and $v_3, v_4 \notin U$. However, v_1, v_2 does not span U, since $(0, 0, 1, 1) \in U$ but cannot be written as a linear combination of v_1 and v_2. Thus, v_1, v_2 is not a basis of U. ■

Exercise 2B9

Suppose $v_1, \dots, v_m$ is a list of vectors in V. For $k \in \{1, \dots, m\}$, let

$$w_k = v_1 + \cdots + v_k.$$

Show that $v_1, \dots, v_m$ is a basis of V if and only if $w_1, \dots, w_m$ is a basis of V.

Solution. By Exercise 2A14, we know that $v_1, \dots, v_m$ is linearly independent if and only if $w_1, \dots, w_m$ is linearly independent. By Exercise 2A3, we know that $v_1, \dots, v_m$ spans V if and only if $w_1, \dots, w_m$ spans V. Thus, $v_1, \dots, v_m$ is a basis of V if and only if $w_1, \dots, w_m$ is a basis of V. ■

Exercise 2B10

Suppose U and W are subspaces of V such that $V = U \oplus W$. Suppose also that $u_1, \dots, u_m$ is a basis of U and $w_1, \dots, w_n$ is a basis of W. Prove that

$$u_1, \dots, u_m, w_1, \dots, w_n$$

is a basis of V.

Solution. Let $u_1, \dots, u_m$ be a basis of U and $w_1, \dots, w_n$ be a basis of W. We claim that the list

$$u_1, \dots, u_m, w_1, \dots, w_n$$

is a basis of V. Let $\alpha_1, \dots, \alpha_m, \beta_1, \dots, \beta_n \in \mathbb{F}$ such that

$$\alpha_1 u_1 + \cdots + \alpha_m u_m + \beta_1 w_1 + \cdots + \beta_n w_n = 0.$$

Then we may write

$$\alpha_1 u_1 + \cdots + \alpha_m u_m = -\beta_1 w_1 - \cdots - \beta_n w_n.$$

Since the left-hand side is in U and the right-hand side is in W, and $U \cap W = \{0\}$, we must have

$$\alpha_1 u_1 + \cdots + \alpha_m u_m = 0 \quad \text{and} \quad \beta_1 w_1 + \cdots + \beta_n w_n = 0.$$

Since $u_1, \dots, u_m$ is a basis of U, we have $\alpha_1 = \cdots = \alpha_m = 0$. Similarly, since $w_1, \dots, w_n$ is a basis of W, we have $\beta_1 = \cdots = \beta_n = 0$. Thus, the list is linearly independent. Moreover, since $V = U \oplus W$, then $v = u + w$ for some $u \in U$ and $w \in W$. Then we may write

$$u = \gamma_1 u_1 + \cdots + \gamma_m u_m \quad \text{and} \quad w = \delta_1 w_1 + \cdots + \delta_n w_n$$

for some $\gamma_1, \ldots, \gamma_m, \delta_1, \ldots, \delta_n \in \mathbb{F}$. Hence,

$$v = u + w = \gamma_1 u_1 + \cdots + \gamma_m u_m + \delta_1 w_1 + \cdots + \delta_n w_n,$$

which shows that the list spans V. Therefore, the list is a basis of V. ■

Exercise 2B11

Suppose V is a real vector space. Show that if $v_1, \ldots, v_n$ is a basis of V (as a real vector space), then $v_1, \ldots, v_n$ is also a basis of the complexification $V_{\mathbb{C}}$ (as a complex vector space).
Note. Please see Exercise 1B8 for the definition of the complexification $V_{\mathbb{C}}$.

Solution. Let $v_1, \ldots, v_n$ be a basis of V. We claim that $v_1, \ldots, v_n$ is also a basis of $V_{\mathbb{C}}$. Let $\alpha_1, \ldots, \alpha_n \in \mathbb{C}$ such that

$$\alpha_1 v_1 + \cdots + \alpha_n v_n = 0.$$

Then we may write each $\alpha_k = a_k + b_k i$ for some $a_k, b_k \in \mathbb{R}$. Then we have

$$(a_1 + b_1 i)v_1 + \cdots + (a_n + b_n i)v_n = 0.$$

This implies that

$$(a_1 v_1 + \cdots + a_n v_n) + i(b_1 v_1 + \cdots + b_n v_n) = 0.$$

Since the left-hand side is a sum of two vectors in V, and V is a real vector space, we must have

$$a_1 v_1 + \cdots + a_n v_n = 0 \quad \text{and} \quad b_1 v_1 + \cdots + b_n v_n = 0.$$

Since $v_1, \ldots, v_n$ is a basis of V, we have $a_1 = \cdots = a_n = 0$ and $b_1 = \cdots = b_n = 0$. Thus, $\alpha_1 = \cdots = \alpha_n = 0$, which shows that the list is linearly independent. Moreover, let $v \in V_{\mathbb{C}}$. Then we may write $v = u + wi$ for some $u, w \in V$. Since $v_1, \ldots, v_n$ is a basis of V, we may write

$$u = \gamma_1 v_1 + \cdots + \gamma_n v_n \quad \text{and} \quad w = \delta_1 v_1 + \cdots + \delta_n v_n$$

for some $\gamma_1, \ldots, \gamma_n, \delta_1, \ldots, \delta_n \in \mathbb{R}$. Hence,

$$v = u + wi = (\gamma_1 + \delta_1 i)v_1 + \cdots + (\gamma_n + \delta_n i)v_n,$$

which shows that the list spans $V_{\mathbb{C}}$. Therefore, the list is a basis of $V_{\mathbb{C}}$. ■

2C Dimension

Exercise 2C1

Show that the subspaces of $\mathbb{R}^2$ are precisely $\{0\}$, all lines in $\mathbb{R}^2$ containing the origin, and $\mathbb{R}^2$.

Solution. Since $\dim(\mathbb{R}^2) = 2$, any subspace of $\mathbb{R}^2$ must have dimension 0, 1, or 2. The only subspace of dimension 0 is $\{0\}$, and the only subspace of dimension 2 is $\mathbb{R}^2$. Any subspace with dimension 1 consists of all scalar multiples of some nonzero $(x, y) \in \mathbb{R}^2$, which is precisely the line with direction (x, y) containing the origin. Thus, the subspaces of $\mathbb{R}^2$ are precisely $\{0\}$, all lines in $\mathbb{R}^2$ containing the origin, and $\mathbb{R}^2$. ■

Exercise 2C2

Show that the subspaces of $\mathbb{R}^3$ are precisely $\{0\}$, all lines in $\mathbb{R}^3$ containing the origin, all planes in $\mathbb{R}^3$ containing the origin, and $\mathbb{R}^3$.

Solution. Since $\dim(\mathbb{R}^3) = 3$, any subspace of $\mathbb{R}^3$ must have dimension 0, 1, 2, or 3. The only subspace of dimension 0 is $\{0\}$, and the only subspace of dimension 3 is $\mathbb{R}^3$. Any subspace with dimension 1 consists of all scalar multiples of some nonzero $(x, y, z) \in \mathbb{R}^3$, which is precisely the line containing the origin in the direction of (x, y, z). Any subspace with dimension 2 consists of all linear combinations of two linearly independent vectors in $\mathbb{R}^3$, which is precisely a plane containing the origin. Thus, the subspaces of $\mathbb{R}^3$ are precisely $\{0\}$, all lines in $\mathbb{R}^3$ containing the origin, all planes in $\mathbb{R}^3$ containing the origin, and $\mathbb{R}^3$. ■

Exercise 2C3

(a) Let $U = \{p \in \mathcal{P}_4(\mathbb{F}) \mid p(6) = 0\}$. Find a basis of U.

(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbb{F})$.

(c) Find a subspace W of $\mathcal{P}_4(\mathbb{F})$ such that $\mathcal{P}_4(\mathbb{F}) = U \oplus W$.

Solution.

(a) Let $p = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 \in U$. Then we have $p(6) = 0$, which gives us the equation

$$a_0 + 6a_1 + 36a_2 + 216a_3 + 1296a_4 = 0.$$

We can solve this equation for a_0 in terms of a_1, a_2, a_3, a_4:

$$a_0 = -6a_1 - 36a_2 - 216a_3 - 1296a_4.$$

Then we can write any polynomial in U as a linear combination of the following four polynomials:

$$u_1 = -6 + x, \quad u_2 = -36 + x^2, \quad u_3 = -216 + x^3, \quad u_4 = -1296 + x^4.$$

Moreover, these four polynomials are linearly independent, since the only way to write the zero polynomial as a linear combination of u_1, u_2, u_3, u_4 is to take all coefficients to be zero. Thus, $\{u_1, u_2, u_3, u_4\}$ is a basis of U.

(b) We can extend the basis $\{u_1, u_2, u_3, u_4\}$ of U to a basis of $\mathcal{P}_4(\mathbb{F})$ by adding one more polynomial that is linearly independent of u_1, u_2, u_3, u_4. For example, we can add the polynomial

$$u_5 = 1,$$

which is linearly independent of u_1, u_2, u_3, u_4 since it is not in U. Thus, $\{u_1, u_2, u_3, u_4, u_5\}$ is a basis of $\mathcal{P}_4(\mathbb{F})$.

(c) Let $W = \operatorname{span} u_5 = \operatorname{span} 1$. Then we have

$$\mathcal{P}_4(\mathbb{F}) = U + W,$$

since every polynomial in $\mathcal{P}_4(\mathbb{F})$ can be uniquely written as a sum of a polynomial in U and a polynomial in W. Moreover, if $p \in U \cap W$, then $p = \alpha$ for some $\alpha \in \mathbb{F}$. Since $p(6) = 0$, we have $\alpha = 0$. Thus, $U \cap W = \{0\}$, and so we have

$$\mathcal{P}_4(\mathbb{F}) = U \oplus W.$$

■

Exercise 2C4

(a) Let $U = \{p \in \mathcal{P}_4(\mathbb{R}) \mid p''(6) = 0\}$. Find a basis of U.

(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbb{R})$.

(c) Find a subspace W of $\mathcal{P}_4(\mathbb{R})$ such that $\mathcal{P}_4(\mathbb{R}) = U \oplus W$.

Solution.

(a) Let $p = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 \in U$. Then we have

$$p''(x) = 2a_2 + 6a_3x + 12a_4x^2,$$

and so $p''(6) = 0$ gives us the equation

$$2a_2 + 36a_3 + 432a_4 = 0.$$

We can solve this equation for a_2 in terms of a_3, a_4:

$$a_2 = -18a_3 - 216a_4.$$

Then we can write any polynomial in U as a linear combination of the following four polynomials:

$$u_1 = 1, \quad u_2 = x, \quad u_3 = -18x^2 + x^3, \quad u_4 = -216x^2 + x^4.$$

Moreover, these four polynomials are linearly independent, since the only way to write the zero polynomial as a linear combination of u_1, u_2, u_3, u_4 is to take all coefficients to be zero. Thus, $\{u_1, u_2, u_3, u_4\}$ is a basis of U.

(b) We can extend the basis $\{u_1, u_2, u_3, u_4\}$ of U to a basis of $\mathcal{P}_4(\mathbb{R})$ by adding one more polynomial that is linearly independent of u_1, u_2, u_3, u_4. For example, we can add the polynomial

$$u_5 = x^2,$$

which is linearly independent of u_1, u_2, u_3, u_4 since it is not in U. Thus, $\{u_1, u_2, u_3, u_4, u_5\}$ is a basis of $\mathcal{P}_4(\mathbb{R})$.

(c) Let $W = \operatorname{span} u_5 = \operatorname{span} x^2$. Then we have

$$\mathcal{P}_4(\mathbb{R}) = U + W,$$

since every polynomial in $\mathcal{P}_4(\mathbb{R})$ can be written as a sum of a polynomial in U and a polynomial in W. Moreover, if $p \in U \cap W$, then $p = \alpha x^2$ for some $\alpha \in \mathbb{R}$. Since $p''(6) = 0$, we have $2\alpha = 0$, which implies that $\alpha = 0$. Thus, $U \cap W = \{0\}$, and so we have

$$\mathcal{P}_4(\mathbb{R}) = U \oplus W.$$

■

Exercise 2C5

(a) Let $U = \{p \in \mathcal{P}_4(\mathbb{F}) \mid p(2) = p(5)\}$. Find a basis of U.

(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbb{F})$.

(c) Find a subspace W of $\mathcal{P}_4(\mathbb{F})$ such that $\mathcal{P}_4(\mathbb{F}) = U \oplus W$.

Solution.

(a) Let $p = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 \in U$. Then we have $p(2) = p(5)$, which gives us the equation

$$a_0 + 2a_1 + 4a_2 + 8a_3 + 16a_4 = a_0 + 5a_1 + 25a_2 + 125a_3 + 625a_4.$$

We can simplify this equation to get

$$-3a_1 - 21a_2 - 117a_3 - 609a_4 = 0.$$

We can solve this equation for a_1 in terms of a_2, a_3, a_4:

$$a_1 = -7a_2 - 39a_3 - 203a_4.$$

Then we can write any polynomial in U as a linear combination of the following four polynomials:

$$u_1 = 1, \quad u_2 = -7x + x^2, \quad u_3 = -39x + x^3, \quad u_4 = -203x + x^4.$$

Moreover, these four polynomials are linearly independent, since the only way to write the zero polynomial as a linear combination of u_1, u_2, u_3, u_4 is to take all coefficients to be zero. Thus, $\{u_1, u_2, u_3, u_4\}$ is a basis of U.

(b) We can extend the basis $\{u_1, u_2, u_3, u_4\}$ of U to a basis of $\mathcal{P}_4(\mathbb{F})$ by adding one more polynomial that is linearly independent of u_1, u_2, u_3, u_4. For example, we can add the polynomial

$$u_5 = x,$$

which is linearly independent of u_1, u_2, u_3, u_4 since it is not in U. Thus, $\{u_1, u_2, u_3, u_4, u_5\}$ is a basis of $\mathcal{P}_4(\mathbb{F})$.

(c) Let $W = \operatorname{span} u_5 = \operatorname{span} x$. Then we have

$$\mathcal{P}_4(\mathbb{F}) = U + W,$$

since every polynomial in $\mathcal{P}_4(\mathbb{F})$ can be written as a sum of a polynomial in U and a polynomial in W. Moreover, if $p \in U \cap W$, then $p = \alpha x$ for some $\alpha \in \mathbb{F}$. Since $p(2) = p(5)$, we have $\alpha 2 = \alpha 5$, which implies that $\alpha = 0$. Thus, $U \cap W = \{0\}$, and so we have

$$\mathcal{P}_4(\mathbb{F}) = U \oplus W. \qquad \blacksquare$$

Exercise 2C6

(a) Let $U = \{p \in \mathcal{P}_4(\mathbb{F}) \mid p(2) = p(5) = p(6)\}$. Find a basis of U.

(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbb{F})$.

(c) Find a subspace W of $\mathcal{P}_4(\mathbb{F})$ such that $\mathcal{P}_4(\mathbb{F}) = U \oplus W$.

Solution.

(a) Let $p = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 \in U$. Then we have $p(2) = p(5) = p(6)$. We know from Exercise 2C5 that the condition $p(2) = p(5)$ gives us the equation

$$a_1 = -7a_2 - 39a_3 - 203a_4.$$

The condition $p(5) = p(6)$ gives us the equation

$$5a_1 + 25a_2 + 125a_3 + 625a_4 = 6a_1 + 36a_2 + 216a_3 + 1296a_4,$$

which simplifies to

$$a_1 + 11a_2 + 91a_3 + 671a_4 = 0.$$

Since we know that $a_1 = -7a_2 - 39a_3 - 203a_4$, we can substitute this into the second equation to get

$$4a_2 + 52a_3 + 468a_4 = 0.$$

We can solve this equation for a_2 in terms of a_3, a_4:

$$a_2 = -13a_3 - 117a_4.$$

Then $a_1 = 52a_3 + 616a_4$. Thus, we can write any polynomial in U as a linear combination of the following three polynomials:

$$u_1 = 1, \quad u_2 = 52x - 13x^2 + x^3, \quad u_3 = 616x - 117x^2 + x^4.$$

Moreover, these three polynomials are linearly independent, since the only way to write the zero polynomial as a linear combination of u_1, u_2, u_3 is to take all coefficients to be zero. Thus, $\{u_1, u_2, u_3\}$ is a basis of U.

(b) We can extend the basis $\{u_1, u_2, u_3\}$ of U to a basis of $\mathcal{P}_4(\mathbb{F})$ by adding two more polynomials that are linearly independent of u_1, u_2, u_3. For example, we can add the polynomials

$$u_4 = x^2, \quad u_5 = x^3,$$

which are linearly independent of u_1, u_2, u_3 since they are not in U. Thus, $\{u_1, u_2, u_3, u_4, u_5\}$ is a basis of $\mathcal{P}_4(\mathbb{F})$.

(c) Let $W = \operatorname{span}(u_4, u_5) = \operatorname{span}(x^2, x^3)$. Then we have

$$\mathcal{P}_4(\mathbb{F}) = U + W,$$

since every polynomial in $\mathcal{P}_4(\mathbb{F})$ can be written as a sum of a polynomial in U and a polynomial in W. Moreover, if $p \in U \cap W$, then $p = \alpha x^2 + \beta x^3$ for some $\alpha, \beta \in \mathbb{F}$. Since $p(2) = p(5) = p(6)$, we have

$$4\alpha + 8\beta = 25\alpha + 125\beta = 36\alpha + 216\beta.$$

Solving this system of equations, we find that $\alpha = \beta = 0$. Thus, $U \cap W = \{0\}$, and so we have

$$\mathcal{P}_4(\mathbb{F}) = U \oplus W. \qquad \blacksquare$$

Exercise 2C7

(a) Let

$$U = \left\{ p \in \mathcal{P}_4(\mathbb{R}) \,\middle|\, \int_{-1}^{1} p = 0 \right\}.$$

Find a basis of U.

(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbb{R})$.

(c) Find a subspace W of $\mathcal{P}_4(\mathbb{R})$ such that $\mathcal{P}_4(\mathbb{R}) = U \oplus W$.

Solution.

(a) Let $p = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 \in U$. Then we have

$$\begin{aligned}\int_{-1}^{1} p &= \int_{-1}^{1} (a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4)\,\mathrm{d}x \\ &= \int_{-1}^{1} (a_0 + a_2x^2 + a_4x^4)\,\mathrm{d}x \\ &= 2\int_{0}^{1} (a_0 + a_2x^2 + a_4x^4)\,\mathrm{d}x \\ &= 2\left(a_0 + \frac{a_2}{3} + \frac{a_4}{5}\right) = 0.\end{aligned}$$

We can solve this equation for a_4 in terms of a_0, a_2:

$$a_4 = -5a_2 - 15a_0.$$

Then we can write any polynomial in U as a linear combination of the following four polynomials:

$$u_1 = 1 - 15x^4, \quad u_2 = x, \quad u_3 = x^2 - 5x^4, \quad u_4 = x^3.$$

Moreover, these four polynomials are linearly independent, since the only way to write the zero polynomial as a linear combination of u_1, u_2, u_3, u_4 is to take all coefficients to be zero. Thus, $\{u_1, u_2, u_3, u_4\}$ is a basis of U.

(b) We can extend the basis $\{u_1, u_2, u_3, u_4\}$ of U to a basis of $\mathcal{P}_4(\mathbb{R})$ by adding one more polynomial that is linearly independent of u_1, u_2, u_3, u_4. For example, we can add the polynomial

$$u_5 = x^4,$$

which is linearly independent of u_1, u_2, u_3, u_4 since it is not in U. Thus, $\{u_1, u_2, u_3, u_4, u_5\}$ is a basis of $\mathcal{P}_4(\mathbb{R})$.

(c) Let $W = \operatorname{span} u_5 = \operatorname{span} x^4$. Then we have

$$\mathcal{P}_4(\mathbb{R}) = U + W,$$

since every polynomial in $\mathcal{P}_4(\mathbb{R})$ can be written as a sum of a polynomial in U and a polynomial in W. Moreover, if $p \in U \cap W$, then $p = \alpha x^4$ for some $\alpha \in \mathbb{R}$. Since the integral of an even function over a symmetric interval is nonzero unless the function is identically zero, we have $\alpha = 0$. Thus, $U \cap W = \{0\}$, and so we have

$$\mathcal{P}_4(\mathbb{R}) = U \oplus W. \qquad \blacksquare$$

Exercise 2C8

Suppose $v_1, \ldots, v_m$ is linearly independent in V and $w \in V$. Prove that

$$\dim \operatorname{span}(v_1 + w, \ldots, v_m + w) \geq m - 1.$$

Solution. Observe that $(v_i + w) - (v_{i+1} + w) = v_i - v_{i+1}$ for $1 \leq i \leq m-1$. Then the list of vectors

$$v_1 - v_2, v_2 - v_3, \ldots, v_{m-1} - v_m$$

is contained in $\operatorname{span}(v_1 + w, \ldots, v_m + w)$. We claim that this list is linearly independent. Suppose that

$$\alpha_1(v_1 - v_2) + \alpha_2(v_2 - v_3) + \cdots + \alpha_{m-1}(v_{m-1} - v_m) = 0$$

for some $\alpha_1, \ldots, \alpha_{m-1} \in \mathbb{F}$. Then we can rewrite this equation as

$$\alpha_1 v_1 + (\alpha_2 - \alpha_1)v_2 + \cdots + (\alpha_{m-1} - \alpha_{m-2})v_{m-1} - \alpha_{m-1}v_m = 0.$$

Since $v_1, \ldots, v_m$ is linearly independent, we must have $\alpha_1 = 0$, $\alpha_2 - \alpha_1 = 0$, $\ldots$, $\alpha_{m-1} - \alpha_{m-2} = 0$, and $-\alpha_{m-1} = 0$. This implies that $\alpha_1 = \cdots = \alpha_{m-1} = 0$. Thus, the list is linearly independent, and so we have

$$\dim \operatorname{span}(v_1 + w, \ldots, v_m + w) \geq m - 1. \qquad \blacksquare$$

Exercise 2C9

Suppose m is a positive integer and $p_0, p_1, \ldots, p_m \in \mathcal{P}(\mathbb{F})$ are such that each p_k has degree k. Prove that $p_0, p_1, \ldots, p_m$ is a basis of $\mathcal{P}_m(\mathbb{F})$.

Solution. Since each p_k has degree k, we can write each polynomial as

$$p_k = a_{k0} + a_{k1}x + \cdots + a_{kk}x^k$$

where $a_{ij} \in \mathbb{F}$ for all $0 \leq i \leq j \leq k$ and $a_{kk} \neq 0$. We claim that $p_0, p_1, \ldots, p_m$ is linearly independent. Suppose that

$$\alpha_0 p_0 + \alpha_1 p_1 + \cdots + \alpha_m p_m = 0$$

for some $\alpha_0, \ldots, \alpha_m \in \mathbb{F}$. Then we can write this equation as

$$(\alpha_0 a_{00} + \alpha_1 a_{10} + \cdots + \alpha_m a_{m0}) + (\alpha_1 a_{11} + \cdots + \alpha_m a_{m1})x + \cdots + (\alpha_m a_{mm})x^m = 0.$$

Since the zero polynomial has all coefficients equal to zero, we must have

$$\alpha_m a_{mm} = 0.$$

Since $a_{mm} \neq 0$, we must have $\alpha_m = 0$. Next, the coefficient of the x^{m-1} term gives us

$$\alpha_{m-1} a_{m-1,m-1} + \alpha_m a_{m,m-1} = 0.$$

Since $\alpha_m = 0$ and $a_{m-1,m-1} \neq 0$, we must have $\alpha_{m-1} = 0$. Continuing in this way, we find that $\alpha_{m-2} = \cdots = \alpha_0 = 0$. Thus, the list is linearly independent. Moreover, since $\dim \mathcal{P}_m(\mathbb{F}) = m + 1$, any linearly independent list of $m + 1$ vectors in $\mathcal{P}_m(\mathbb{F})$ is a basis. Therefore, $p_0, p_1, \ldots, p_m$ is a basis of $\mathcal{P}_m(\mathbb{F})$. $\blacksquare$

Exercise 2C10

Suppose m is a positive integer. For $0 \leq k \leq m$, let

$$p_k(x) = x^k(1 - x)^{m-k}.$$

Show that $p_0, \ldots, p_m$ is a basis of $\mathcal{P}_m(\mathbb{F})$.

Solution. We first determine what exponents appear in the polynomial p_k. Using the Binomial Theorem, we can expand $(1 - x)^{m-k}$ and obtain the expression for $p_k(x)$ as a polynomial in x:

$$p_k(x) = x^k \sum_{i=0}^{m-k} \binom{m-k}{i} (-1)^i x^i = \sum_{i=0}^{m-k} \binom{m-k}{i} (-1)^i x^{k+i}.$$

Thus, the exponents that appear in p_k are $k, k + 1, \ldots, m$, and we may write p_k as

$$p_k(x) = a_{kk}x^k + a_{k,k+1}x^{k+1} + \cdots + a_{km}x^m$$

where $a_{kk} = 1$ and $a_{kj} \in \mathbb{F}$ for all $k \leq j \leq m$. We claim that $p_0, p_1, \ldots, p_m$ is linearly independent. Suppose that

$$\alpha_0 p_0 + \alpha_1 p_1 + \cdots + \alpha_m p_m = 0$$

for some $\alpha_0, \ldots, \alpha_m \in \mathbb{F}$. Then we can write this equation as

$$(\alpha_0 a_{00}) + (\alpha_0 a_{01} + \alpha_1 a_{11})x + \cdots + (\alpha_0 a_{0m} + \cdots + \alpha_m a_{mm})x^m = 0.$$

Since the zero polynomial has all coefficients equal to zero, and $a_{00} \neq 0$ by definition of p_0, we must have $\alpha_0 = 0$. Next, the coefficient of the x is $\alpha_0 a_{01} + \alpha_1 a_{11} = 0$. Since $\alpha_0 = 0$ and $a_{11} \neq 0$, we must have $\alpha_1 = 0$. Continuing in this way, we find that $\alpha_2 = \cdots = \alpha_m = 0$. Thus, the list is linearly independent. Moreover, since $\dim \mathcal{P}_m(\mathbb{F}) = m+1$, any linearly independent list of $m+1$ vectors in $\mathcal{P}_m(\mathbb{F})$ is a basis. Therefore, $p_0, p_1, \ldots, p_m$ is a basis of $\mathcal{P}_m(\mathbb{F})$. ■

Exercise 2C11

Suppose U and W are both four-dimensional subspaces of $\mathbb{C}^6$. Prove that there exist two vectors in $U \cap W$ such that neither of these vectors is a scalar multiple of the other.

Solution. Let U and W be four-dimensional subspaces of $\mathbb{C}^6$. By the dimension formula for the sum of two subspaces, we have

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W) = 4 + 4 - \dim(U \cap W) = 8 - \dim(U \cap W).$$

Since $U + W$ is a subspace of $\mathbb{C}^6$, we have $\dim(U + W) \leq 6$. Thus, we have

$$8 - \dim(U \cap W) \leq 6,$$

which implies that $\dim(U \cap W) \geq 2$. Therefore, there exist at least two linearly independent vectors in $U \cap W$, and neither of these vectors is a scalar multiple of the other. ■

Exercise 2C12

Suppose that U and W are subspaces of $\mathbb{R}^8$ such that $\dim U = 3$, $\dim W = 5$, and $U + W = \mathbb{R}^8$. Prove that $\mathbb{R}^8 = U \oplus W$.

Solution. By the dimension formula for the sum of two subspaces, we have

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W) = 3 + 5 - \dim(U \cap W) = 8 - \dim(U \cap W).$$

Since $U + W = \mathbb{R}^8$, we have $\dim(U + W) = 8$. Thus, we have

$$8 - \dim(U \cap W) = 8,$$

which implies that $\dim(U \cap W) = 0$. Therefore, $U \cap W = \{0\}$, and so we have

$$\mathbb{R}^8 = U \oplus W.$$

■

Exercise 2C13

Suppose U and W are both five-dimensional subspaces of $\mathbb{R}^8$. Prove that $U \cap W \neq \{0\}$.

Solution. By the dimension formula for the sum of two subspaces, we have

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W) = 5 + 5 - \dim(U \cap W) = 10 - \dim(U \cap W).$$

Since $U + W$ is a subspace of $\mathbb{R}^8$, we have $\dim(U + W) \leq 8$. Thus, we have

$$10 - \dim(U \cap W) \leq 8,$$

which implies that $\dim(U \cap W) \geq 2$. Therefore, $U \cap W$ is nontrivial, and so we have $U \cap W \neq \{0\}$. ■

Exercise 2C14

Suppose V is a ten-dimensional vector space and V_1, V_2, V_3 are subspaces of V with $\dim V_1 = \dim V_2 = \dim V_3 = 7$. Prove that $V_1 \cap V_2 \cap V_3 \neq \{0\}$.

Solution. Let $W = V_1 \cap V_2$. By the dimension formula for the sum of two subspaces, we have

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim W = 7 + 7 - \dim W = 14 - \dim W.$$

Since $V_1 + V_2$ is a subspace of V, we have $\dim(V_1 + V_2) \leq 10$. Thus, we have $14 - \dim W \leq 10$, which implies that $\dim W \geq 4$. Therefore, $V_1 \cap V_2$ is at least four-dimensional. Now, consider the subspace $W \cap V_3 = V_1 \cap V_2 \cap V_3$. By the dimension formula for the sum of two subspaces, we have

$$\dim(W + V_3) = \dim W + \dim V_3 - \dim(W \cap V_3) = \dim W + 7 - \dim(W \cap V_3).$$

Since $W + V_3$ is a subspace of V, we have $\dim(W + V_3) \leq 10$. Thus, we have

$$\dim W + 7 - \dim(W \cap V_3) \leq 10,$$

which implies that $\dim(W \cap V_3) \geq \dim W + 7 - 10 = \dim W - 3$. Since $\dim W \geq 4$, we have $\dim(W \cap V_3) \geq 1$. Therefore, $V_1 \cap V_2 \cap V_3$ is nontrivial, and so we have $V_1 \cap V_2 \cap V_3 \neq \{0\}$. ■

Exercise 2C15

Suppose V is finite-dimensional and V_1, V_2, V_3 are subspaces of V with $\dim V_1 + \dim V_2 + \dim V_3 > 2 \dim V$. Prove that $V_1 \cap V_2 \cap V_3 \neq \{0\}$.

Solution. Let $W = V_1 \cap V_2$. By the dimension formula for the sum of two subspaces, we have

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim W.$$

Since $V_1 + V_2$ is a subspace of V, we have $\dim(V_1 + V_2) \leq \dim V$. Thus, we have

$$\dim V_1 + \dim V_2 - \dim W \leq \dim V,$$

which implies that $\dim W \geq \dim V_1 + \dim V_2 - \dim V$. Now, consider the subspace $W \cap V_3 = V_1 \cap V_2 \cap V_3$. By the dimension formula for the sum of two subspaces, we have

$$\dim(W + V_3) = \dim W + \dim V_3 - \dim(W \cap V_3).$$

Since $W + V_3$ is a subspace of V, we have $\dim(W + V_3) \leq \dim V$. Thus, we have

$$\dim W + \dim V_3 - \dim(W \cap V_3) \leq \dim V,$$

which implies that $\dim(W \cap V_3) \geq \dim W + \dim V_3 - \dim V$. Since $\dim W \geq \dim V_1 + \dim V_2 - \dim V$, we have

$$\dim(W \cap V_3) \geq (\dim V_1 + \dim V_2 - \dim V) + \dim V_3 - \dim V = \dim V_1 + \dim V_2 + \dim V_3 - 2 \dim V.$$

Since $\dim V_1 + \dim V_2 + \dim V_3 > 2 \dim V$, we have $\dim(W \cap V_3) > 0$. Therefore, $V_1 \cap V_2 \cap V_3$ is nontrivial, and so we have $V_1 \cap V_2 \cap V_3 \neq \{0\}$. ■

Exercise 2C16

Suppose V is finite-dimensional and U is a subspace of V with $U \neq V$. Let $n = \dim V$ and $m = \dim U$. Prove that there exist $n - m$ subspaces of V, each of dimension $n - 1$, whose intersection equals U.

Solution. Let U be a subspace of V with $\dim U = m < n = \dim V$. We can extend a basis of U to a basis of V. Let $\{u_1, \ldots, u_m\}$ be a basis of U, and extend it to a basis of V by adding vectors $v_1, \ldots, v_{n-m}$. Then $\{u_1, \ldots, u_m, v_1, \ldots, v_{n-m}\}$ is a basis of V. For each $i = 1, \ldots, n - m$, let

$$W_i = \operatorname{span}(u_1, \ldots, u_m, v_1, \ldots, v_{i-1}, v_{i+1}, \ldots, v_{n-m}).$$

Then each W_i is a subspace of V with dimension $n-1$, since it is spanned by $n-1$ linearly independent vectors. Moreover, we have

$$U = W_1 \cap W_2 \cap \cdots \cap W_{n-m},$$

since any vector in the intersection must be a linear combination of the basis vectors of U, and any vector in U is contained in all the subspaces W_i. Therefore, there exist $n - m$ subspaces of V, each of dimension $n - 1$, whose intersection equals U. ■

Exercise 2C17

Suppose that $V_1, \ldots, V_m$ are finite-dimensional subspaces of V. Prove that $V_1 + \cdots + V_m$ is finite-dimensional and

$$\dim(V_1 + \cdots + V_m) \leq \dim V_1 + \cdots + \dim V_m.$$

Solution. We will prove the statement by induction on m. For the base case, let $m = 2$. Then we have

$$V_1 + V_2 = V_1 + V_2,$$

which is finite-dimensional and

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2) \leq \dim V_1 + \dim V_2.$$

Hence, the statement holds for $m = 2$. Now, suppose that the statement holds for some positive integer m. That is, suppose that if $V_1, \ldots, V_m$ are finite-dimensional subspaces of V, then $V_1 + \cdots + V_m$ is finite-dimensional and

$$\dim(V_1 + \cdots + V_m) \leq \dim V_1 + \cdots + \dim V_m.$$

We want to show that the statement holds for $m + 1$. Let $V_1, \ldots, V_{m+1}$ be finite-dimensional subspaces of V. By the induction hypothesis, we know that $W = V_1 + \cdots + V_m$ is finite-dimensional and

$$\dim W \leq \dim V_1 + \cdots + \dim V_m.$$

Then we have

$$\dim(W + V_{m+1}) = \dim W + \dim V_{m+1} - \dim(W \cap V_{m+1}).$$

Since $W \cap V_{m+1}$ is a subspace of V, we have $\dim(W \cap V_{m+1}) \geq 0$. Thus, we have

$$\dim(W + V_{m+1}) \leq \dim W + \dim V_{m+1} \leq \dim V_1 + \cdots + \dim V_m + \dim V_{m+1}.$$

Therefore, $V_1 + \cdots + V_{m+1}$ is finite-dimensional and

$$\dim(V_1 + \cdots + V_{m+1}) \leq \dim V_1 + \cdots + \dim V_m + \dim V_{m+1}.$$

■

Exercise 2C18

Suppose V is finite-dimensional, with $\dim V = n \geq 1$. Prove that there exist one-dimensional subspaces $V_1, \ldots, V_n$ of V such that
$$V = V_1 \oplus \cdots \oplus V_n.$$

Solution. Let $v_1, \ldots, v_n$ be a basis of V. For each $i = 1, \ldots, n$, let $V_i = \operatorname{span} v_i$. Then each V_i is a one-dimensional subspace of V, since it is spanned by a single nonzero vector. Moreover, we have

$$V = V_1 + \cdots + V_n,$$

and the sum is direct because the basis vectors are linearly independent. Therefore, we have

$$V = V_1 \oplus \cdots \oplus V_n. \qquad \blacksquare$$

Exercise 2C19

Explain why you might guess, motivated by analogy with the formula for the number of elements in the union of three finite sets, that if V_1, V_2, V_3 are subspaces of a finite-dimensional vector space, then

$$\begin{aligned}\dim(V_1 + V_2 + V_3) = {} & \dim V_1 + \dim V_2 + \dim V_3 \\ & - \dim(V_1 \cap V_2) - \dim(V_1 \cap V_3) - \dim(V_2 \cap V_3) \\ & + \dim(V_1 \cap V_2 \cap V_3).\end{aligned}$$

Then either prove the formula above or give a counterexample.

Solution. One may guess that the given formula above is true by analogy with the formula for the number of elements in the union of three finite sets. However, this erroneously conflates that the idea of subspace sums is equivalent to the idea of set unions. In fact, the formula is not true in general. For example, let $V = \mathbb{R}^2$, and let
$$V_1 = \operatorname{span}((1,0)), \quad V_2 = \operatorname{span}((0,1)), \quad V_3 = \operatorname{span}((1,1)).$$
It is easy to see that $\dim V_1 = \dim V_2 = \dim V_3 = 1$, but $\dim(V_1 \cap V_2) = \dim(V_1 \cap V_3) = \dim(V_2 \cap V_3) = 0$ and $\dim(V_1 \cap V_2 \cap V_3) = 0$. Thus, the right-hand side of the formula is

$$1 + 1 + 1 - 0 - 0 - 0 + 0 = 3,$$

but the left-hand side is $\dim(V_1 + V_2 + V_3) = \dim \mathbb{R}^2 = 2$. Therefore, the formula is not true in general. $\blacksquare$

Exercise 2C20

Prove that if V_1, V_2, and V_3 are subspaces of a finite-dimensional vector space, then

$$\begin{aligned}\dim(V_1 + V_2 + V_3) = {} & \dim V_1 + \dim V_2 + \dim V_3 \\ & - \frac{\dim(V_1 \cap V_2) + \dim(V_1 \cap V_3) + \dim(V_2 \cap V_3)}{3} \\ & - \frac{\dim((V_1 + V_2) \cap V_3) + \dim((V_1 + V_3) \cap V_2) + \dim((V_2 + V_3) \cap V_1)}{3}.\end{aligned}$$

Solution. Note that we need to account for all pairs of subspaces all pairs of sums of subspaces, since the sum of subspaces is not equivalent to the union of sets. We can use the dimension formula repeatedly to obtain the answer. To that end, define the following subspaces:

$$W_1 = V_1 + V_2, \quad W_2 = V_1 + V_3, \quad W_3 = V_2 + V_3.$$

Applying the dimension formula to each of W_1, W_2, W_3, we have

$$\begin{aligned}
\dim W_1 &= \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2), \\
\dim W_2 &= \dim V_1 + \dim V_3 - \dim(V_1 \cap V_3), \\
\dim W_3 &= \dim V_2 + \dim V_3 - \dim(V_2 \cap V_3).
\end{aligned}$$

Next, we can apply the dimension formula to each of the sums $W_1 + V_3, W_2 + V_2, W_3 + V_1$:

$$\begin{aligned}
\dim(W_1 + V_3) &= \dim W_1 + \dim V_3 - \dim(W_1 \cap V_3), \\
&= \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2) + \dim V_3 - \dim(W_1 \cap V_3) \\
\dim(W_2 + V_2) &= \dim W_2 + \dim V_2 - \dim(W_2 \cap V_2), \\
&= \dim V_1 + \dim V_3 - \dim(V_1 \cap V_3) + \dim V_2 - \dim(W_2 \cap V_2) \\
\dim(W_3 + V_1) &= \dim W_3 + \dim V_1 - \dim(W_3 \cap V_1), \\
&= \dim V_2 + \dim V_3 - \dim(V_2 \cap V_3) + \dim V_1 - \dim(W_3 \cap V_1).
\end{aligned}$$

Note that $W_1 + V_3 = W_2 + V_2 = W_3 + V_1 = V_1 + V_2 + V_3$. Thus, we can add the three equations above and divide by 3 to obtain the desired formula:

$$\begin{aligned}
\dim(V_1 + V_2 + V_3) = {} & \dim V_1 + \dim V_2 + \dim V_3 \\
& - \frac{\dim(V_1 \cap V_2) + \dim(V_1 \cap V_3) + \dim(V_2 \cap V_3)}{3} \\
& - \frac{\dim((V_1 + V_2) \cap V_3) + \dim((V_1 + V_3) \cap V_2) + \dim((V_2 + V_3) \cap V_1)}{3}.
\end{aligned}$$

■